



Research paper

Closed-form forward kinematics of a novel class of 3-RRR spherical parallel mechanisms with coplanar intermediate-joint axes and coplanar distal-joint axes

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ABSTRACT

The forward kinematics (FK) of 3-RRR spherical parallel mechanisms (SPMs) has been studied extensively, which is essential for their calibration and control. These studies indicate that most 3-RRR SPMs are associated with octic characteristic equations, which must be solved using iterative approaches. This paper introduces a new class of 3-RRR SPMs featuring simplified FK formulations, characterized by three coplanar intermediate-joint axes and three coplanar distal-joint axes. Unlike most 3-RRR SPMs, this class admits only up to four distinct assembly modes. Moreover, four distinct formulations are proposed, all yielding quartic characteristic equations, thereby enabling closed-form FK solutions. Furthermore, one special subclass of the robot class is found to admit quadratic characteristic equations with simpler coefficient expressions. Beyond simplified FK relations, this robot class also offers unlimited rotation capability about certain directions, making it potentially attractive for practical applications. It is worth noting that this robot class provides flexibility in selecting architecture parameters, thereby allowing for a wide range of designs. Additionally, the proposed formulations can also be applied to certain parallel mechanisms with alternative topologies.

1. Introduction

When compared to serial mechanisms, parallel mechanisms generally offer higher accuracy, rigidity, load-carrying capacity, and superior dynamic performance, which have received considerable research attention in recent decades [1–7]. Among them, 3-RRR spherical parallel mechanisms (SPMs) have been extensively studied and find applications in various fields, such as three-degree-of-freedom (six-DOF) orienting devices, robotic shoulders and wrists, haptic devices, medical devices, spherical motors, etc. [8–13].

The forward kinematics (FK) involves determining the pose of the moving platform (MP) for a given set of input joint variables, which is crucial for the simulation, calibration and feedback control of parallel mechanisms [14,15]. General 3-RRR SPMs have been extensively studied by various researchers using different methods, resulting in octic characteristic equations at best [16–18]. This suggests that general 3-RRR SPMs can admit up to eight sets of distinct FK solutions, corresponding to eight distinct assembly modes. Consequently, numerical methods relying on iterative procedures are needed in most cases, which may suffer from drawbacks such as high computational cost, convergence to incorrect solution branches, and even convergence failure [14]. On the

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other hand, 3-RRR SPMs with closed-form forward kinematics solutions (CFFKS) offer advantages due to their simplified FK relations and readily available symbolic solutions, making them potentially appealing for practical applications [19]. These attributes not only streamline tasks such as simulation, calibration, and feedback control but also facilitate trajectory planning within the input joint space, including minimal-energy trajectory planning [14].

Mechanisms with CFFKS, sometimes referred to as “analytic manipulators”, are advantageous in practical applications from the kinematics perspective, which have been extensively investigated by researchers [20–24]. For instance, various researchers have devoted to the development of three-DOF planar parallel mechanisms with CFFKS, leading to the identification of a relatively large set of such mechanisms upon leveraging special architecture parameters [19,25–27]. Similar efforts have been made for other types of parallel mechanisms, such as those with a topology of 2UPS-UP [24], 3-SPS-S [28], 3-UPU-S [29], 3-CRR [30], or 3-RRPS [31]. As for the case of 3-RRR SPMs, the most well-known example with CFFKS is the Agile Eye [22,32], comprising an orthogonal base platform (BP), an orthogonal MP, three 90° proximal links and three 90° distal links. Notably, the Agile Eye has been shown to admit characteristic equations that are quadratic [22] or linear [32]. Another example (that admits quartic characteristic equation) was identified during the study of a 3-CCC parallel mechanism, which consists of an orthogonal BP, a coplanar MP, three 90° proximal links, and three 90° distal links [33]. Moreover, a class of 3-RRR SPMs with CFFKS was identified by Huang et al. [34], encompassing all 3-RRR SPMs for which two of the three distal-joint axes are coincident, although these mechanisms architectures are bound to be non-symmetric. More recently, three classes of 3-RRR SPMs with CFFKS were proposed, each featuring three 90° distal links; moreover, the first two classes have either an orthogonal MP or a coplanar MP, while the third class features three coaxial input joint axes and three coplanar intermediate joint axes [35].

In this paper, a novel class of 3-RRR SPMs with CFFKS, characterized by three coplanar intermediate-joint axes and three coplanar distal-joint axes, is investigated. Interestingly, the maximum number of distinct FK solutions or, equivalently, the maximum number of distinct assembly modes, is identified, which is found to be four, in contrast to the general case of eight. Moreover, four distinct formulations are proposed for the FK of the robot class, all yielding quartic characteristic equations. Furthermore, a special subclass of the robot class is investigated, resulting in quadratic characteristic equations with simpler coefficient expressions. It is worth noting that the three input-joint axes are arranged to be coaxial for this robot class, a layout that has been utilized in a large number of PMs [36–39]. Such a layout has been shown to yield a number of advantages, such as unlimited rotation capability about certain directions, reduced limb interference, and generally larger workspace volumes. These characteristics render the robot class potentially attractive for practical applications. Notably, five of the architecture parameters can be freely specified for the robot class, indicating that the robot class encompasses a large set of potential designs. Moreover, the proposed formulations are also applicable to a number of PMs with alternative topologies—see Section 3.6 for more details.

The paper is organized as follows: Section 2 introduces the architecture of the robot class under study, along with the relevant notations. In Section 3, four different formulations for the FK of the robot class are presented, followed by an examination of their simplifications for a special subclass of the robot class, and a discussion on their potential applications to certain PMs with alternative topologies. Section 4 provides several case studies to compare these formulations. Finally, Section 5 concludes the paper.

2. Architecture description

Fig. 1(a) illustrates an example of the 3-RRR SPM under study, whose MP is connected to the BP via three limbs, each limb containing three serially-connected revolute joints. In order to describe its geometry, we introduce the following notations, as shown in Fig. 1(b): The unit vectors parallel to the proximal, intermediate and distal R joint axes of the i th limb are denoted as $\mathbf{u}_i$, $\mathbf{v}_i$ and $\mathbf{w}_i$, respectively, for $i = 1, 2, 3$. Moreover, the angle between $\mathbf{u}_i$ and $\mathbf{v}_i$, henceforth referred to as the dimension of the i th proximal link, is denoted as α_i ; similarly, the angle between $\mathbf{v}_i$ and $\mathbf{w}_i$, henceforth referred to as the dimension of the i th distal link, is denoted as β_i ; Then, the robot class under study is characterized by the following conditions:

1. The axes of the three proximal R joints are all coincident, i.e., $\mathbf{u}_1 = \mathbf{u}_2 = \mathbf{u}_3$;
2. The dimensions of the three proximal links are all 90°, i.e., $\alpha_i = 90^\circ$, for $i = 1, 2, 3$.
3. The three distal-joint axes lie in a common plane (denoted as Π_w hereafter), with none of them being parallel to each other.¹
This condition can be characterized by $m_1 \mathbf{w}_1 + m_2 \mathbf{w}_2 + m_3 \mathbf{w}_3 = \mathbf{0}$, with none of $\{m_i\}_1^3$ vanishing.

Without loss of generality, we establish the BP frame $\mathcal{F}_B$ such that its origin O coincides with the rotating center of the MP, with the negative direction of its z -axis aligned with $\{\mathbf{u}_i\}_1^3$. Moreover, the x -axis can be arbitrarily selected within the plane normal to the z -axis, as shown in Fig. 1(b). Apparently, $\{\mathbf{v}_i\}_1^3$ also always lie in a common plane (the x - O - y plane), which will be denoted as Π_v hereafter. Moreover, we denote the unit normal vectors of Π_v and Π_w as $\mathbf{n}_v$ and $\mathbf{n}_w$, respectively, to facilitate the description of their orientations.

Next, we introduce one more set of architecture parameters and two more sets of configuration parameters: Let γ_i denote the angle from $\mathbf{w}_{i+1}$ to $\mathbf{w}_{i+2}$ (measured about $\mathbf{n}_w$), for $i = 1, 2, 3$. Interestingly, $\{\gamma_i\}_1^3$ can be used to generate a valid set of $\{m_i\}_1^3$, namely, $\{m_i = \sin \gamma_i\}_1^3$. Moreover, we denote the array of actuated joint variables as $\boldsymbol{\theta} = [\theta_1, \theta_2, \theta_3]^T$, where θ_i is the angle from the plane formed by the z -axis and the x -axis to the plane of the i th proximal link, measured about the direction of $\mathbf{u}_i$. Lastly, ϵ_i denotes the angle from the plane of the i th proximal link to that of the i th distal link, measured about the direction of $\mathbf{v}_i$, for $i = 1, 2, 3$.

¹ Notice that we have excluded the case where two of the distal-joint axes are parallel. In such a case, the 3-RRR SPM under study can be decomposed into two 5R SPMs, whose FK has been thoroughly addressed in previous literature [34,40–42].

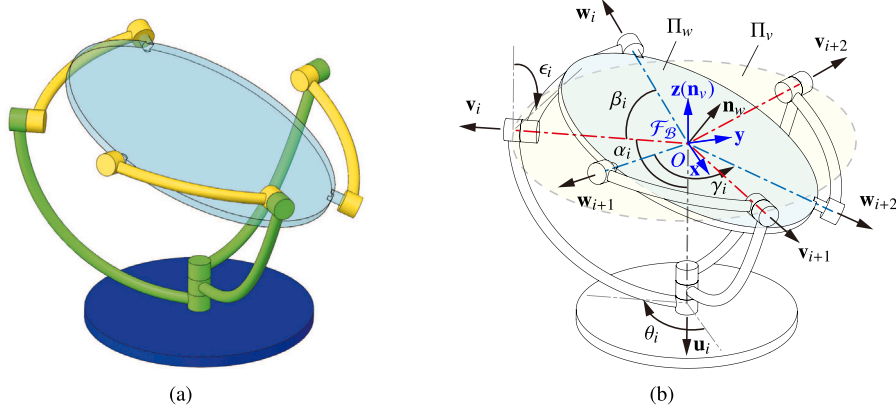


Fig. 1. (a) An example of the 3-RRR SPM under study in an arbitrary posture, and (b) its corresponding notation.

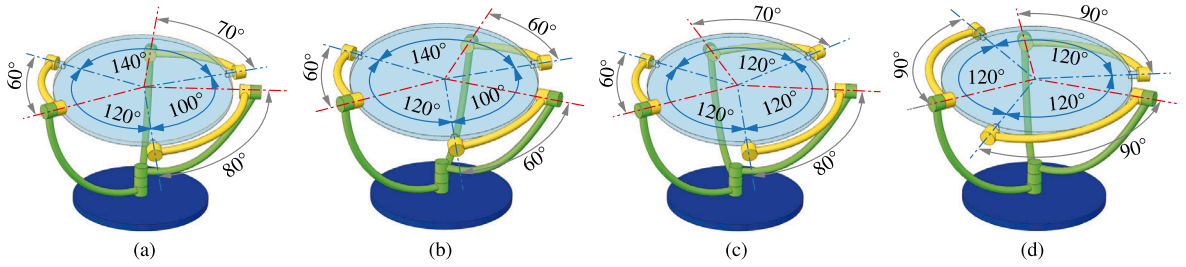


Fig. 2. Four design examples of 3-RRR SPMs from the robot class under study.

Table 1
The architecture parameters of the design examples shown in Fig. 2.

Design example	β_1	β_2	β_3	γ_1	γ_2	γ_3	m_1	m_2	m_3
I	60°	70°	80°	100°	120°	140°	$\sin 100^\circ$	$\sqrt{3}/2$	$\sin 140^\circ$
II	60°	60°	60°	100°	120°	140°	$\sin 100^\circ$	$\sqrt{3}/2$	$\sin 140^\circ$
III	60°	70°	80°	120°	120°	120°	$\sqrt{3}/2$	$\sqrt{3}/2$	$\sqrt{3}/2$
IV	90°	90°	90°	120°	120°	120°	$\sqrt{3}/2$	$\sqrt{3}/2$	$\sqrt{3}/2$

As can be seen, the shape of the BP and the dimensions of the proximal links are the same for all 3-RRR SPMs within the robot class under study. However, the dimensions of the three distal links ($\{\beta_i\}_1^3$) can be freely specified. Similarly, two of $\{\gamma_i\}_1^3$ can be arbitrarily specified, while the remaining one can be determined based on the coplanarity condition for $\{\mathbf{w}_i\}_1^3$. As a result, five architecture parameters can be freely chosen for this robot class, thereby encompassing a large set of potential designs. Several design examples belonging to this robot class are illustrated in Fig. 2, whose architecture parameters, together with $\{m_i\}_1^3$, are listed in Table 1. Notice that the first example represents a general design with five arbitrarily chosen design parameters ($\{\beta_i\}_1^3$ and $\{\gamma_i\}_1^3$), while the second and third examples represent two designs exhibiting certain special geometries—one featuring three distal links of equal dimensions and the other with three evenly distributed distal-link axes (on plane Π_w). Moreover, the fourth example represents a special design belonging to the proposed robot class, which, as will be shown latter, yields an even simpler characteristic equation.

Finally, it is worth mentioning that the three input joint axes are arranged to be coaxial in order to guarantee that the three intermediate-joint axes are always coplanar. Such a coaxial layout for the input joint axes has been utilized in various 3-RRR SPMs [8,37,38,43,44], which brings about additional benefits, such as unlimited rotation capacity about certain directions, significantly reduced interference among different limbs, and generally larger workspace volumes [8,37]. Moreover, the actuation of the three coaxial input R joints can be realized using different strategies [36–39]. For example, a common approach is to use a belt transmission system [36] or a gear transmission system [37] for each actuated R joint, as illustrated in Fig. 3(a) and (b). Another common approach is to implement each actuated R joint as a prismatic slider moving along a circular guide [38,39], as shown in Fig. 3(c). For various examples of these strategies and their physical implementations, see the following Refs. [36–39].

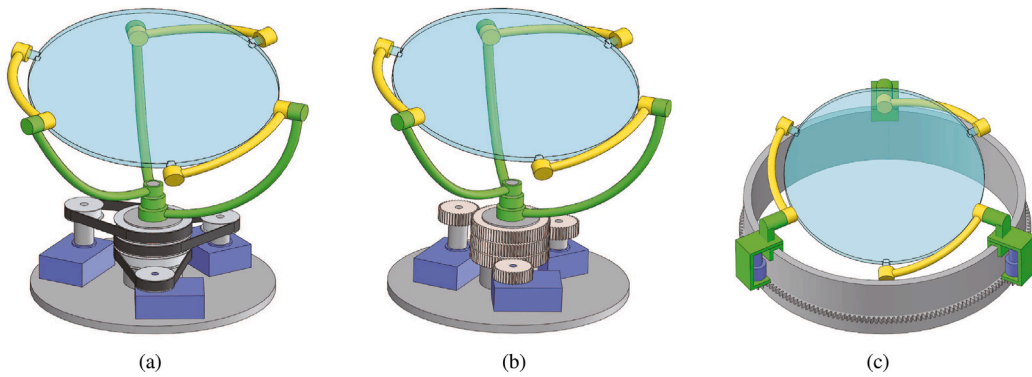


Fig. 3. Three possible actuation strategies for the three coaxial input R joints using (a) belt transmission systems, (b) gear transmission systems, and (c) prismatic sliders on a circular guide.

3. Forward kinematics of the robot class under study

Apparently, $\{\mathbf{u}_i\}_1^3$ and $\{\mathbf{v}_i\}_1^3$ are all readily available in the FK problem. Consequently, all we need is to determine the directions of $\{\mathbf{w}_i\}_1^3$, after which the MP orientation can be readily determined. In this section, the FK of the robot class under study is conducted using four different formulations, all of which lead to quartic characteristic equations. Moreover, a special subclass is investigated thereafter, where the aforementioned formulations can be further simplified, leading to quadratic characteristic equations with simpler coefficient expressions.

3.1. The first formulation for general 3-RRR SPMs of the robot class under study

In this formulation, $\{\epsilon_i\}_1^3$, the rotation angles of the three intermediate joints, are used as the unknowns; moreover, three constraint equations are established, after which two of the unknowns are eliminated, leading to a quartic equation in $\sin \epsilon_1$.

As stated earlier, the coplanarity of $\{\mathbf{w}_i\}_1^3$ can be characterized by the following condition:

$$m_1 \mathbf{w}_1 + m_2 \mathbf{w}_2 + m_3 \mathbf{w}_3 = \mathbf{0} \quad (1)$$

where $m_i = \sin \gamma_i$ is readily available, for $i = 1, 2, 3$. Next, we establish three constraint equations based on Eq. (1). To do this, we first decompose $\mathbf{w}_i$ along three mutually orthogonal directions, namely,

$$\mathbf{w}_i = \sin \beta_i \cos \epsilon_i \mathbf{n}_v + \sin \beta_i \sin \epsilon_i \mathbf{v}_{i\perp} + \cos \beta_i \mathbf{v}_i, \quad i = 1, 2, 3 \quad (2)$$

where $\mathbf{v}_{i\perp} \equiv \mathbf{v}_i \times \mathbf{n}_v$, for $i = 1, 2, 3$. Upon plugging Eq. (2) into Eq. (1), we obtain

$$\sum_{i=1}^3 (m_i \sin \beta_i \cos \epsilon_i) \mathbf{n}_v + \sum_{i=1}^3 (m_i \sin \beta_i \sin \epsilon_i \mathbf{v}_{i\perp}) + \sum_{i=1}^3 (m_i \cos \beta_i \mathbf{v}_i) = \mathbf{0} \quad (3)$$

Moreover, upon dot-multiplying both sides of Eq. (3) with $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_3$ and $\mathbf{n}_v$, four scalar equations in terms of $\{\epsilon_i\}_1^3$ are obtained, namely,

$$\begin{aligned} m_{2s} \sin \theta_{21} \sin \epsilon_2 + m_{3s} \sin \theta_{31} \sin \epsilon_3 + \sum_{i=1}^3 m_{ic} \cos \theta_{i1} &= 0 \\ m_{1s} \sin \theta_{12} \sin \epsilon_1 + m_{3s} \sin \theta_{32} \sin \epsilon_3 + \sum_{i=1}^3 m_{ic} \cos \theta_{i2} &= 0 \\ m_{1s} \sin \theta_{13} \sin \epsilon_1 + m_{2s} \sin \theta_{23} \sin \epsilon_2 + \sum_{i=1}^3 m_{ic} \cos \theta_{i3} &= 0 \\ m_{1s} \cos \epsilon_1 + m_{2s} \cos \epsilon_2 + m_{3s} \cos \epsilon_3 &= 0 \end{aligned} \quad (4)$$

where $m_{ic} \equiv m_i \cos \beta_i$, $m_{is} \equiv m_i \sin \beta_i$ and $\theta_{ij} \equiv \theta_j - \theta_i$ are all readily available, for $i, j = 1, 2, 3$. It is noteworthy that the first three of Eqs. (4) are always linearly dependent, making one of them redundant. As a result, we will proceed with the following derivations based on the last three of Eqs. (4). However, it will be shown shortly that the first of Eqs. (4) will be needed later on to address the special case where $\mathbf{v}_2 \parallel \mathbf{v}_3$.

Now, let us address the case where $\mathbf{v}_2 \nparallel \mathbf{v}_3$ first, which occurs when $\theta_{23} \neq k\pi$, $k \in \mathbb{Z}$. In this scenario, a univariate equation can be obtained by eliminating ϵ_2 and ϵ_3 from the last three of Eqs. (4). To achieve this, we rewrite the second and third of Eqs. (4) in the following form:

$$\begin{aligned} \sin \epsilon_2 &= -(m_{1s} \sin \theta_{13} \sin \epsilon_1 + \sum_{i=1}^3 m_{ic} \cos \theta_{i3}) / (m_{2s} \sin \theta_{23}) \equiv \eta_1 \sin \epsilon_1 + \eta_2 \\ \sin \epsilon_3 &= -(m_{1s} \sin \theta_{12} \sin \epsilon_1 + \sum_{i=1}^3 m_{ic} \cos \theta_{i2}) / (m_{3s} \sin \theta_{32}) \equiv \eta_3 \sin \epsilon_1 + \eta_4 \end{aligned} \quad (5)$$

where $\{\eta_i\}_1^4$ are intermediate variables that can be calculated as

$$\begin{aligned}\eta_1 &\equiv -m_{1s} \sin \theta_{13} / (m_{2s} \sin \theta_{23}), & \eta_2 &\equiv -\sum_{i=1}^3 (m_{ic} \cos \theta_{i3}) / (m_{2s} \sin \theta_{23}) \\ \eta_3 &\equiv -m_{1s} \sin \theta_{12} / (m_{3s} \sin \theta_{32}), & \eta_4 &\equiv -\sum_{i=1}^3 (m_{ic} \cos \theta_{i2}) / (m_{3s} \sin \theta_{32})\end{aligned}\quad (6)$$

Moreover, the last of Eqs. (4) can be rewritten as

$$m_{2s} \cos \epsilon_2 + m_{3s} \cos \epsilon_3 = -m_{1s} \cos \epsilon_1 \quad (7)$$

After squaring both sides of Eq. (7) and then replacing $\cos^2 \epsilon_i$ with $1 - \sin^2 \epsilon_i$, for $i = 1, 2, 3$, we obtain

$$m_{2s}^2 \sin^2 \epsilon_2 + m_{3s}^2 \sin^2 \epsilon_3 - m_{1s}^2 \sin^2 \epsilon_1 + (m_{1s}^2 - m_{2s}^2 - m_{3s}^2) = 2m_{2s}m_{3s} \cos \epsilon_2 \cos \epsilon_3 \quad (8)$$

Moreover, $\sin \epsilon_2$ and $\sin \epsilon_3$ can be eliminated upon inserting Eqs. (5) into Eq. (8). After straightforward manipulations, the following constraint equation is obtained, namely,

$$\eta_5 \sin^2 \epsilon_1 + \eta_6 \sin \epsilon_1 + \eta_7 = \eta_8 \cos \epsilon_2 \cos \epsilon_3 \quad (9)$$

where

$$\begin{aligned}\eta_5 &\equiv m_{1s}^2 (\sin^2 \theta_{12} + \sin^2 \theta_{13} - \sin^2 \theta_{23}) \\ \eta_6 &\equiv m_{1c} m_{1s} (\sin 2\theta_{12} + \sin 2\theta_{13}) + 2m_{1s} m_{2c} (\sin \theta_{12} + \sin \theta_{13} \cos \theta_{23}) + 2m_{1s} m_{3c} (\sin \theta_{13} + \sin \theta_{12} \cos \theta_{23}) \\ \eta_7 &\equiv m_{1c}^2 (\cos^2 \theta_{12} + \cos^2 \theta_{13}) + (m_{1s}^2 - m_{2s}^2 - m_{3s}^2) \sin^2 \theta_{23} + 4m_{2c} m_{3c} \cos \theta_{23} + 2m_{2c}^2 + 2m_{3c}^2 \\ &\quad + 2m_{1c} m_{2c} (\cos \theta_{12} + \cos \theta_{13} \cos \theta_{23}) + 2m_{1c} m_{3c} (\cos \theta_{13} + \cos \theta_{12} \cos \theta_{23}) \\ \eta_8 &\equiv 2m_{2s} m_{3s} \sin^2 \theta_{23}\end{aligned}\quad (10)$$

Notice that a common factor $m_{2s}^2 m_{3s}^2 \sin^2 \theta_{23}$ arises in all four coefficients of Eq. (9), which has been eliminated in the expressions of $\{\eta_i\}_5^8$ for the sake of simplicity. Moreover, if we further square both sides of Eq. (9), replace $\cos^2 \epsilon_i$ with $1 - \sin^2 \epsilon_i$, for $i = 2, 3$, and then plug Eqs. (5) in, a univariate equation in $\sin \epsilon_1$ can be obtained, namely,

$$\zeta_4 \sin^4 \epsilon_1 + \zeta_3 \sin^3 \epsilon_1 + \zeta_2 \sin^2 \epsilon_1 + \zeta_1 \sin \epsilon_1 + \zeta_0 = 0 \quad (11)$$

with

$$\begin{aligned}\zeta_4 &\equiv \eta_5^2 - \eta_1^2 \eta_3^2 \eta_8^2, & \zeta_3 &\equiv 2\eta_5 \eta_6 - 2(\eta_2 \eta_3 + \eta_1 \eta_4) \eta_1 \eta_3 \eta_8^2 \\ \zeta_2 &\equiv \eta_6^2 + 2\eta_5 \eta_7 + \eta_8^2 ((1 - \eta_4^2) \eta_1^2 + (1 - \eta_2^2) \eta_3^2 - 4\eta_2 \eta_3 \eta_4 \eta_1) \\ \zeta_1 &\equiv 2\eta_6 \eta_7 + 2\eta_8^2 ((1 - \eta_4^2) \eta_1 \eta_2 + (1 - \eta_2^2) \eta_3 \eta_4), & \zeta_0 &\equiv \eta_7^2 - (1 - \eta_2^2)(1 - \eta_4^2) \eta_8^2\end{aligned}\quad (12)$$

which is the quartic characteristic equation sought. Apparently, such a quartic equation theoretically admits up to four distinct solutions for $\sin \epsilon_1$. However, as will be shown in Appendix A, there are at most two distinct valid solutions among these four, which, in turn, leads to up to four valid solutions for ϵ_1 . Moreover, for each ϵ_1 thus obtained, a unique set of $\{\sin \epsilon_2, \sin \epsilon_3\}$ can be determined from the second and third of Eqs. (4), after which a unique set of $\{\cos \epsilon_2, \cos \epsilon_3\}$ can be determined based on the following two equations:

$$\begin{aligned}\mathbf{w}_1 \cdot \mathbf{w}_2 &= (\sin \beta_2 \mathbf{w}_1 \cdot \mathbf{n}_v) \cos \epsilon_2 + (\sin \beta_2 \mathbf{w}_1 \cdot \mathbf{v}_{2\perp}) \sin \epsilon_2 + \cos \beta_2 \mathbf{w}_1 \cdot \mathbf{v}_2 = \cos \gamma_3 \\ \mathbf{w}_1 \cdot \mathbf{w}_3 &= (\sin \beta_3 \mathbf{w}_1 \cdot \mathbf{n}_v) \cos \epsilon_3 + (\sin \beta_3 \mathbf{w}_1 \cdot \mathbf{v}_{3\perp}) \sin \epsilon_3 + \cos \beta_3 \mathbf{w}_1 \cdot \mathbf{v}_3 = \cos \gamma_2\end{aligned}\quad (13)$$

To this end, it is apparent that each valid ϵ_1 corresponds to a unique set of $\{\epsilon_2, \epsilon_3\}$, implying that general 3-RRR SPMs of the robot class under study admit up to four distinct sets of FK solutions or, equivalently, four distinct assembly modes.

It is worth noting that, although the aforementioned formulation leads to a characteristic equation in $\sin \epsilon_1$, similar characteristic equations in $\sin \epsilon_2$ or $\sin \epsilon_3$ can be obtained by simply exchanging subscripts 1, 2, 3 with 2, 3, 1 or 3, 1, 2 in the aforementioned formulation. These alternative characteristic equations can be used when Eq. (11) fails to produce the correct solutions for $\sin \epsilon_1$.

Next, we address the special case where $\mathbf{v}_2 \parallel \mathbf{v}_3$, which occurs when $\theta_{23} = k\pi$, $k \in \mathbb{Z}$. It can be readily verified that the second and third of Eqs. (4) become linearly dependent in this case, ultimately leading to the vanishing of all the coefficients in Eq. (11). As a result, the aforementioned formulation is no longer applicable in this case. However, this special case can be readily handled simply upon utilizing the characteristic equations in $\sin \epsilon_2$ or $\sin \epsilon_3$ instead, as described earlier. Alternatively, this special case can be solved via the following procedure: Since $\sin \theta_{23} = 0$ in this situation, the second of Eqs. (4) becomes a univariate equation in $\sin \epsilon_1$, from which two values of ϵ_1 can be obtained; moreover, for each ϵ_1 thus obtained, the corresponding $\mathbf{w}_1$ can be readily determined, after which two possible values of ϵ_2 can be calculated based on the following relation:

$$\mathbf{w}_1 \cdot \mathbf{w}_2 = (\sin \beta_2 \mathbf{w}_1 \cdot \mathbf{n}_v) \cos \epsilon_2 + (\sin \beta_2 \mathbf{w}_1 \cdot \mathbf{v}_{2\perp}) \sin \epsilon_2 + \cos \beta_2 \mathbf{w}_1 \cdot \mathbf{v}_2 = \cos \gamma_3 \quad (14)$$

Finally, $\mathbf{w}_3$ can be calculated based on Eq. (1). Apparently, this special case also admits up to four sets of FK solutions.

3.2. The second formulation for general 3-RRR SPMs of the robot class under study

Next, we present an alternative formulation for the FK problem of the proposed robot class, which also uses the rotation angles of the three intermediate joints as unknowns. As will be shown later, the characteristic equations derived from the first formulation and this new formulation are very similar, except that their corresponding coefficients differ by a common factor. This difference makes the new formulation more robust under certain robot configurations.

Now, let us introduce the second formulation: Firstly, we recall that $\mathbf{w}_i$ can be decomposed along three mutually orthogonal directions, namely,

$$\mathbf{w}_i = \sin \beta_i \cos \epsilon_i \mathbf{n}_v + \sin \beta_i \sin \epsilon_i \mathbf{v}_{1\perp} + \cos \beta_i \mathbf{v}_i, \quad i = 1, 2, 3 \quad (15)$$

Then, a constraint equation in ϵ_1 and ϵ_2 can be established upon exploiting the relation between $\mathbf{w}_1$ and $\mathbf{w}_2$, namely,

$$\begin{aligned} \mathbf{w}_1 \cdot \mathbf{w}_2 = \sin \beta_1 \sin \beta_2 \cos \epsilon_1 \cos \epsilon_2 + \sin \beta_1 \sin \beta_2 \sin \epsilon_1 \sin \epsilon_2 \mathbf{v}_{1\perp} \cdot \mathbf{v}_{2\perp} + \sin \beta_1 \cos \beta_2 \sin \epsilon_1 \mathbf{v}_{1\perp} \cdot \mathbf{v}_2 \\ + \cos \beta_1 \sin \beta_2 \sin \epsilon_2 \mathbf{v}_1 \cdot \mathbf{v}_{2\perp} + \cos \beta_1 \cos \beta_2 \mathbf{v}_1 \cdot \mathbf{v}_2 = \cos \gamma_3 \end{aligned} \quad (16)$$

Moreover, considering the condition $m_1 \mathbf{w}_1 + m_2 \mathbf{w}_2 + m_3 \mathbf{w}_3 = 0$, a second constraint equation in ϵ_1 and ϵ_2 can be obtained upon dot-multiplying its both sides by $\mathbf{v}_3$, namely,

$$\begin{aligned} m_1 \mathbf{v}_3 \cdot \mathbf{w}_1 + m_2 \mathbf{v}_3 \cdot \mathbf{w}_2 + m_3 \cos \beta_3 = m_1 \sin \beta_1 \sin \epsilon_1 \mathbf{v}_{1\perp} \cdot \mathbf{v}_3 + m_1 \cos \beta_1 \mathbf{v}_1 \cdot \mathbf{v}_3 \\ + m_2 \sin \beta_2 \sin \epsilon_2 \mathbf{v}_{2\perp} \cdot \mathbf{v}_3 + m_2 \cos \beta_2 \mathbf{v}_2 \cdot \mathbf{v}_3 + m_3 \cos \beta_3 = 0 \end{aligned} \quad (17)$$

Next, we rearrange Eqs. (16) and (17) into the following form:

$$\begin{aligned} E_1 \cos \epsilon_2 + F_1 \sin \epsilon_2 = G_1 \\ F_2 \sin \epsilon_2 = G_2 \end{aligned} \quad (18)$$

where

$$E_1 \equiv \delta_1 \cos \epsilon_1, \quad F_1 \equiv \delta_2 + \delta_3 \sin \epsilon_1, \quad G_1 \equiv \delta_4 + \delta_5 \sin \epsilon_1, \quad F_2 \equiv \delta_6, \quad G_2 \equiv \delta_7 + \delta_8 \sin \epsilon_1 \quad (19)$$

with

$$\begin{aligned} \delta_1 &\equiv \sin \beta_1 \sin \beta_2, \quad \delta_2 \equiv -\cos \beta_1 \sin \beta_2 \sin \theta_{12}, \quad \delta_3 \equiv \sin \beta_1 \sin \beta_2 \cos \theta_{12}, \quad \delta_4 \equiv \cos \gamma_3 - \cos \beta_1 \cos \beta_2 \cos \theta_{12} \\ \delta_5 &\equiv -\sin \beta_1 \cos \beta_2 \sin \theta_{12}, \quad \delta_6 \equiv -m_2 \sin \beta_2 \sin \theta_{23}, \quad \delta_7 \equiv m_1 \cos \beta_1 \cos \theta_{13} + m_2 \cos \beta_2 \cos \theta_{23} + m_3 \cos \beta_3 \\ \delta_8 &\equiv m_1 \sin \beta_1 \sin \theta_{13} \end{aligned} \quad (20)$$

and $\theta_{ij} \equiv \theta_j - \theta_i$. Upon solving for $\cos \epsilon_2$ and $\sin \epsilon_2$ from Eq. (18) and substituting them into the identity $\cos^2 \epsilon_2 + \sin^2 \epsilon_2 = 1$, a univariate equation in $\sin \epsilon_1$ is obtained, which, after straightforward manipulations, becomes

$$\zeta'_4 \sin^4 \epsilon_1 + \zeta'_3 \sin^3 \epsilon_1 + \zeta'_2 \sin^2 \epsilon_1 + \zeta'_1 \sin \epsilon_1 + \zeta'_0 = 0 \quad (21)$$

where

$$\begin{aligned} \zeta'_4 &\equiv (\delta_3^2 - \delta_1^2) \delta_8^2, \quad \zeta'_3 \equiv 2(\delta_3^2 - \delta_1^2) \delta_7 \delta_8 + 2\delta_3 \delta_8 (\delta_2 \delta_8 - \delta_5 \delta_6) \\ \zeta'_2 &\equiv (\delta_6^2 - \delta_7^2) \delta_1^2 + (\delta_1^2 + \delta_2^2) \delta_8^2 + (\delta_3 \delta_7 - \delta_5 \delta_6)^2 + 2\delta_2 \delta_8 (\delta_3 \delta_7 - \delta_5 \delta_6) + 2\delta_3 \delta_8 (\delta_2 \delta_7 - \delta_4 \delta_6) \\ \zeta'_1 &\equiv 2(\delta_1^2 + \delta_2^2) \delta_7 \delta_8 + 2(\delta_2 \delta_7 - \delta_4 \delta_6) (\delta_3 \delta_7 - \delta_5 \delta_6) - 2\delta_2 \delta_4 \delta_6 \delta_8, \quad \zeta'_0 \equiv -(\delta_6^2 - \delta_7^2) \delta_1^2 + (\delta_2 \delta_7 - \delta_4 \delta_6)^2 \end{aligned} \quad (22)$$

It is noteworthy that the characteristic equations in the first and second formulations, as shown in Eqs. (11) and (21), are both in terms of $\sin \epsilon_1$; moreover, it has been verified that the coefficients of these two characteristic equations are very similar, except that they differ by a common factor, namely,

$$\zeta'_i = \sin^2 \beta_2 / (4m_1^2 \sin^2 \theta_{23}) \zeta_i, \quad i = 0, 1, 2, 3, 4 \quad (23)$$

To this end, up to four distinct solutions for $\sin \epsilon_1$ can be obtained from Eq. (21), resulting in up to eight distinct solutions for ϵ_1 . Once ϵ_1 is determined, ϵ_2 can be found using Eq. (18), after which it becomes a simple matter to determine the directions of $\{\mathbf{w}_i\}_1^3$. It is noteworthy that at most four of these eight sets of FK solutions are valid, as made apparent by Appendix A. Moreover, unlike the case for the first formulation, the second formulation remains applicable even when $\theta_{23} = k\pi$, $k \in \mathbb{Z}$, making it generally more robust.

3.3. The third formulation for general 3-RRR SPMs of the robot class under study

During the investigation of the first two formulations, an interesting observation emerged regarding the angle between the unit normal vectors of planes Π_v and Π_w . Specifically, for all robot postures corresponding to a given set of input joint variables (as obtained in the forward kinematics problem), the angle between $\mathbf{n}_v$ and $\mathbf{n}_w$ takes at most two distinct values. This observation prompted us to explore alternative formulations for the forward kinematics, resulting in the development of the third and fourth formulations.

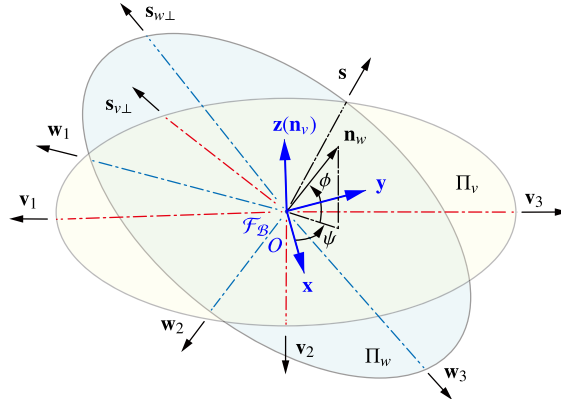


Fig. 4. Illustration of the unit direction vectors relevant to the third and fourth formulations.

Before describing the third formulation, we introduce the following notations, as shown in Fig. 4: Let $\mathbf{s}$ denote the unit vector parallel to the intersection line of Π_v and Π_w ; moreover, $\mathbf{s}_{v\perp}$ denotes the unit vector in Π_v that is orthogonal to $\mathbf{s}$, while $\mathbf{s}_{w\perp}$ denotes the unit vector in Π_w that is orthogonal to $\mathbf{s}$. Without loss of generality, we choose their directions such that

$$\mathbf{s} = (\mathbf{n}_v \times \mathbf{n}_w) / \|\mathbf{n}_v \times \mathbf{n}_w\|_2, \quad \mathbf{s}_{v\perp} = \mathbf{n}_v \times \mathbf{s}, \quad \mathbf{s}_{w\perp} = \mathbf{n}_w \times \mathbf{s} \quad (24)$$

Then, each of the two sets of vectors $\{\mathbf{n}_v, \mathbf{s}, \mathbf{s}_{v\perp}\}$ and $\{\mathbf{n}_w, \mathbf{s}, \mathbf{s}_{w\perp}\}$ forms a right-handed orthonormal triad. Moreover, $\{\mathbf{v}_i\}_1^3$ and $\{\mathbf{w}_i\}_1^3$ can be decomposed as

$$\mathbf{v}_i = \cos \vartheta_{vi} \mathbf{s} + \sin \vartheta_{vi} \mathbf{s}_{v\perp}, \quad \mathbf{w}_i = \cos \vartheta_{wi} \mathbf{s} + \sin \vartheta_{wi} \mathbf{s}_{w\perp} \quad (25)$$

where ϑ_{vi} denotes the angle from $\mathbf{s}$ to $\mathbf{v}_i$, measured about the direction of $\mathbf{n}_v$ (or equivalently, $-\mathbf{u}_i$), and ϑ_{wi} denotes the angle from $\mathbf{s}$ to $\mathbf{w}_i$, measured about the direction of $\mathbf{n}_w$, for $i = 1, 2, 3$. Moreover,

$$\vartheta_{v2} - \vartheta_{v1} = -\theta_{12}, \quad \vartheta_{v3} - \vartheta_{v1} = -\theta_{13}, \quad \vartheta_{w2} - \vartheta_{w1} = \gamma_3, \quad \vartheta_{w3} - \vartheta_{w1} = -\gamma_2 \quad (26)$$

where θ_{12} , θ_{13} , γ_2 and γ_3 are all available in the FK problem. This means that ϑ_{v2} , ϑ_{v3} , ϑ_{w2} and ϑ_{w3} can all be expressed as linear functions of ϑ_{v1} or ϑ_{w1} ; this observation will be shown to be useful in the following derivations.

Now, three constraint equations can be established based on the following relations:

$$\mathbf{v}_i \cdot \mathbf{w}_i = \cos \beta_i, \quad \text{for } i = 1, 2, 3 \quad (27)$$

In order to facilitate their computation, we introduce the coordinates of $\mathbf{n}_v$ and $\mathbf{n}_w$ in the base frame $\mathcal{F}_B$, namely,

$$[\mathbf{n}_v]_B = [0, 0, 1]^T, \quad [\mathbf{n}_w]_B = [\cos \psi \cos \phi, \sin \psi \cos \phi, \sin \phi]^T \quad (28)$$

where $\psi \in [-\pi, \pi)$ and $\phi \in [-\pi/2, \pi/2]$ are as yet to be determined. Then, the coordinates of $\mathbf{s}$, $\mathbf{s}_{v\perp}$ and $\mathbf{s}_{w\perp}$ in $\mathcal{F}_B$ can be readily determined as

$$[\mathbf{s}]_B = [-\sin \psi, \cos \psi, 0]^T, \quad [\mathbf{s}_{v\perp}]_B = [-\cos \psi, -\sin \psi, 0]^T, \quad [\mathbf{s}_{w\perp}]_B = [-\cos \psi \sin \phi, -\sin \psi \sin \phi, \cos \phi]^T \quad (29)$$

Upon substituting Eqs. (25) and (29) into Eq. (27), we can calculate it as

$$\sin \phi \sin \vartheta_{wi} \sin \vartheta_{vi} + \cos \vartheta_{wi} \cos \vartheta_{vi} = \cos \beta_i, \quad \text{for } i = 1, 2, 3 \quad (30)$$

Moreover, we define ϑ_{si} and ϑ_{mi} such that

$$\vartheta_{si} \equiv (\vartheta_{vi} + \vartheta_{wi}), \quad \vartheta_{mi} \equiv (\vartheta_{vi} - \vartheta_{wi}), \quad \text{for } i = 1, 2, 3 \quad (31)$$

Then, Eq. (30) can be transformed into the following form:

$$\sin \phi (\cos \vartheta_{mi} - \cos \vartheta_{si}) + (\cos \vartheta_{mi} + \cos \vartheta_{si}) = 2 \cos \beta_i, \quad \text{for } i = 1, 2, 3 \quad (32)$$

Next, we eliminate ϕ from Eqs. (32). To do this, we rewrite Eqs. (32) as

$$\sin \phi = \frac{2 \cos \beta_i - (\cos \vartheta_{mi} + \cos \vartheta_{si})}{(\cos \vartheta_{mi} - \cos \vartheta_{si})}, \quad \text{for } i = 1, 2, 3 \quad (33)$$

Then, two constraint equations without ϕ can be obtained, namely,

$$\begin{aligned} \frac{2 \cos \beta_1 - (\cos \vartheta_{m1} + \cos \vartheta_{s1})}{(\cos \vartheta_{m1} - \cos \vartheta_{s1})} &= \frac{2 \cos \beta_2 - (\cos \vartheta_{m2} + \cos \vartheta_{s2})}{(\cos \vartheta_{m2} - \cos \vartheta_{s2})} \\ \frac{2 \cos \beta_1 - (\cos \vartheta_{m1} + \cos \vartheta_{s1})}{(\cos \vartheta_{m1} - \cos \vartheta_{s1})} &= \frac{2 \cos \beta_3 - (\cos \vartheta_{m3} + \cos \vartheta_{s3})}{(\cos \vartheta_{m3} - \cos \vartheta_{s3})} \end{aligned} \quad (34)$$

Apparently, Eqs. (34) involve two equations with six unknowns ($\{\vartheta_{si}\}_1^3$ and $\{\vartheta_{mi}\}_1^3$). However, four of them ($\{\vartheta_{si}\}_2^3$ and $\{\vartheta_{mi}\}_2^3$) can be readily eliminated upon resorting to the following relations:

$$\begin{aligned}\vartheta_{s2} - \vartheta_{s1} &= (\vartheta_{v2} + \vartheta_{w2}) - (\vartheta_{v1} + \vartheta_{w1}) = (\vartheta_{v2} - \vartheta_{v1}) + (\vartheta_{w2} - \vartheta_{w1}) = -\theta_{12} + \gamma_3 \equiv \sigma_{12} \\ \vartheta_{s3} - \vartheta_{s1} &= (\vartheta_{v3} + \vartheta_{w3}) - (\vartheta_{v1} + \vartheta_{w1}) = (\vartheta_{v3} - \vartheta_{v1}) + (\vartheta_{w3} - \vartheta_{w1}) = -\theta_{13} - \gamma_2 \equiv \sigma_{13} \\ \vartheta_{m2} - \vartheta_{m1} &= (\vartheta_{v2} - \vartheta_{w2}) - (\vartheta_{v1} - \vartheta_{w1}) = (\vartheta_{v2} - \vartheta_{v1}) - (\vartheta_{w2} - \vartheta_{w1}) = -\theta_{12} - \gamma_3 \equiv \mu_{12} \\ \vartheta_{m3} - \vartheta_{m1} &= (\vartheta_{v3} - \vartheta_{w3}) - (\vartheta_{v1} - \vartheta_{w1}) = (\vartheta_{v3} - \vartheta_{v1}) - (\vartheta_{w3} - \vartheta_{w1}) = -\theta_{13} + \gamma_2 \equiv \mu_{13}\end{aligned}\quad (35)$$

where σ_{12} , σ_{13} , μ_{12} and μ_{13} are readily available in the FK problem. Upon plugging Eqs. (35) into Eqs. (34), ϑ_{s2} , ϑ_{s3} , ϑ_{m2} and ϑ_{m3} can be eliminated, leading to a system of two equations in ϑ_{s1} and ϑ_{m1} , namely,

$$\begin{aligned}E_1 \cos \vartheta_{m1} + F_1 \sin \vartheta_{m1} &= G_1 \\ E_2 \cos \vartheta_{m1} + F_2 \sin \vartheta_{m1} &= G_2\end{aligned}\quad (36)$$

with

$$\begin{aligned}E_1 &\equiv 2 \cos \beta_2 - 2 \cos \beta_1 \cos \mu_{12} + (2 \cos \mu_{12} - 2 \cos \sigma_{12}) \cos \vartheta_{s1} + 2 \sin \sigma_{12} \sin \vartheta_{s1} \\ F_1 &\equiv 2 \sin \mu_{12} (\cos \beta_1 - \cos \vartheta_{s1}), \quad G_1 \equiv (2 \cos \beta_2 - 2 \cos \beta_1 \cos \sigma_{12}) \cos \vartheta_{s1} + 2 \cos \beta_1 \sin \sigma_{12} \sin \vartheta_{s1} \\ E_2 &\equiv 2 \cos \beta_3 - 2 \cos \beta_1 \cos \mu_{13} + (2 \cos \mu_{13} - 2 \cos \sigma_{13}) \cos \vartheta_{s1} + 2 \sin \sigma_{13} \sin \vartheta_{s1} \\ F_2 &\equiv 2 \sin \mu_{13} (\cos \beta_1 - \cos \vartheta_{s1}), \quad G_2 \equiv (2 \cos \beta_3 - 2 \cos \beta_1 \cos \sigma_{13}) \cos \vartheta_{s1} + 2 \cos \beta_1 \sin \sigma_{13} \sin \vartheta_{s1}\end{aligned}\quad (37)$$

Next, $\cos \vartheta_{m1}$ and $\sin \vartheta_{m1}$ can be calculated from Eq. (36), namely,

$$\cos \vartheta_{m1} = \frac{\lambda_1 \cos \vartheta_{s1} + \lambda_2 \sin \vartheta_{s1}}{\lambda_5 \cos \vartheta_{s1} + \lambda_6 \sin \vartheta_{s1} + \lambda_7}, \quad \sin \vartheta_{m1} = \frac{\lambda_3 \cos \vartheta_{s1} + \lambda_4 \sin \vartheta_{s1}}{\lambda_5 \cos \vartheta_{s1} + \lambda_6 \sin \vartheta_{s1} + \lambda_7}\quad (38)$$

with

$$\begin{aligned}\lambda_1 &\equiv -\cos \beta_3 \sin \mu_{12} + \cos \beta_2 \sin \mu_{13} + \cos \beta_1 (\sin \mu_{12} \cos \sigma_{13} - \sin \mu_{13} \cos \sigma_{12}) \\ \lambda_2 &\equiv \cos \beta_1 (\sin \mu_{13} \sin \sigma_{12} - \sin \mu_{12} \sin \sigma_{13}) \\ \lambda_3 &\equiv \cos \beta_3 (\cos \sigma_{12} - \cos \mu_{12}) + \cos \beta_2 (\cos \mu_{13} - \cos \sigma_{13}) + \cos \beta_1 (\cos \mu_{12} \cos \sigma_{13} - \cos \mu_{13} \cos \sigma_{12}) \\ \lambda_4 &\equiv -\cos \beta_3 \sin \sigma_{12} + \cos \beta_2 \sin \sigma_{13} + \cos \beta_1 (\cos \mu_{13} \sin \sigma_{12} - \cos \mu_{12} \sin \sigma_{13}) \\ \lambda_5 &\equiv -\sin \mu_{12} \cos \mu_{13} + \sin \mu_{12} \cos \sigma_{13} + \sin \mu_{13} (\cos \mu_{12} - \cos \sigma_{12}) \\ \lambda_6 &\equiv \sin \mu_{13} \sin \sigma_{12} - \sin \mu_{12} \sin \sigma_{13}, \quad \lambda_7 \equiv -\cos \beta_3 \sin \mu_{12} + \cos \beta_1 \sin (\mu_{12} - \mu_{13}) + \cos \beta_2 \sin \mu_{13}\end{aligned}\quad (39)$$

Notice that a common factor $4(\cos \beta_1 - \cos \vartheta_{s1})$ arises in all the coefficients in Eq. (38), which has been eliminated in the expressions of $\{\lambda_i\}_1^7$ for the sake of simplicity. To this end, a univariate equation in ϑ_{s1} can be obtained upon inserting Eq. (38) into the constraint equation $\cos^2 \vartheta_{m1} + \sin^2 \vartheta_{m1} = 1$, namely,

$$\begin{aligned}(\lambda_1^2 + \lambda_3^2 - \lambda_5^2) \cos^2 \vartheta_{s1} + (\lambda_2^2 + \lambda_4^2 - \lambda_6^2) \sin^2 \vartheta_{s1} + (2\lambda_1\lambda_2 + 2\lambda_3\lambda_4 - 2\lambda_5\lambda_6) \cos \vartheta_{s1} \sin \vartheta_{s1} \\ - 2\lambda_5\lambda_7 \cos \vartheta_{s1} - 2\lambda_6\lambda_7 \sin \vartheta_{s1} - \lambda_7^2 = 0\end{aligned}\quad (40)$$

Moreover, upon introducing $T \equiv \tan(\vartheta_{s1}/2)$, both $\sin \vartheta_{s1}$ and $\cos \vartheta_{s1}$ can be expressed as functions of T , namely,

$$\cos \vartheta_{s1} = \frac{1 - T^2}{1 + T^2}, \quad \sin \vartheta_{s1} = \frac{2T}{1 + T^2}\quad (41)$$

After plugging Eq. (41) into Eq. (40) and some straightforward manipulations, a quartic univariate equation in T is obtained, namely,

$$\zeta_4 T^4 + \zeta_3 T^3 + \zeta_2 T^2 + \zeta_1 T + \zeta_0 = 0\quad (42)$$

with

$$\begin{aligned}\zeta_4 &\equiv \lambda_1^2 + \lambda_3^2 - (\lambda_5 - \lambda_7)^2, \quad \zeta_3 \equiv -4(\lambda_1\lambda_2 + \lambda_3\lambda_4 + \lambda_6(\lambda_7 - \lambda_5)) \\ \zeta_2 &\equiv -2(\lambda_1^2 - 2\lambda_2^2 + \lambda_3^2 - 2\lambda_4^2 - \lambda_5^2 + 2\lambda_6^2 + \lambda_7^2) \\ \zeta_1 &\equiv 4(\lambda_1\lambda_2 + \lambda_3\lambda_4 - \lambda_6(\lambda_5 + \lambda_7)), \quad \zeta_0 \equiv \lambda_1^2 + \lambda_3^2 - (\lambda_5 + \lambda_7)^2\end{aligned}\quad (43)$$

It is noteworthy that the expressions of $\{\lambda_i\}_1^7$, as displayed in Eq. (39), can be converted to an alternative form, namely, $\Lambda = \mathbf{K}\Theta$, where $\Lambda = [\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5, \lambda_6, \lambda_7]^T$, matrix $\mathbf{K}$ is a function of $\{\beta_i\}_1^3$ and $\{\gamma_i\}_1^3$ that can be calculated beforehand, and Θ is an 8×1 array that only involves the input joint variables $\{\theta_i\}_1^3$. Since $\mathbf{K}$ is a constant matrix for a given robot that only needs to be calculated once, this alternative form leads to a more efficient routine for the computation of Λ —see Appendix B for the relevant expressions.

As can be seen, the characteristic equation shown in Eq. (42) is quartic, which admits up to four distinct solutions for T . Moreover, for each T thus obtained, a unique set of $\cos \vartheta_{s1}$, $\sin \vartheta_{s1}$, $\cos \vartheta_{m1}$, $\sin \vartheta_{m1}$, $\sin \phi$ and ϕ can be determined based on Eqs. (41), (38), (33) and the condition $\phi \in [-\pi/2, \pi/2]$. From these, the solutions for ϑ_{s1} and ϑ_{m1} are found as $\vartheta_{s1} = \tilde{\vartheta}_{s1} + 2k_1\pi$ and $\vartheta_{m1} = \tilde{\vartheta}_{m1} + 2k_2\pi$, with $\tilde{\vartheta}_{s1}$ and $\tilde{\vartheta}_{m1}$ representing their solutions within the range $[0, 2\pi)$, and $k_1, k_2 \in \mathbb{Z}$. To this end, the corresponding ϑ_{v1} and ϑ_{w1} can be calculated as

$$\begin{aligned}\vartheta_{v1} &= (\vartheta_{s1} + \vartheta_{m1})/2 = (\tilde{\vartheta}_{s1} + \tilde{\vartheta}_{m1})/2 + (k_1 + k_2)\pi \\ \vartheta_{w1} &= (\vartheta_{s1} - \vartheta_{m1})/2 = (\tilde{\vartheta}_{s1} - \tilde{\vartheta}_{m1})/2 + (k_1 - k_2)\pi\end{aligned}\quad (44)$$

where $(k_1 + k_2)$ and $(k_1 - k_2)$ represent two numbers that are both even or both odd (because their difference $2k_1$ is even). Considering that adding multiples of 2π to ϑ_{v1} or ϑ_{w1} does not affect the directions of $\mathbf{v}_1$ and $\mathbf{w}_1$, the solutions shown in Eq. (44) can be reduced to the following two set of solutions:

$$\vartheta_{v1} = (\tilde{\vartheta}_{s1} + \tilde{\vartheta}_{m1})/2, \vartheta_{w1} = (\tilde{\vartheta}_{s1} - \tilde{\vartheta}_{m1})/2, \quad \text{or} \quad \vartheta_{v1} = (\tilde{\vartheta}_{s1} + \tilde{\vartheta}_{m1})/2 + \pi, \vartheta_{w1} = (\tilde{\vartheta}_{s1} - \tilde{\vartheta}_{m1})/2 + \pi \quad (45)$$

This means that each solution for T corresponds to two different sets of ϑ_{v1} and ϑ_{w1} . Moreover, for each set of ϑ_{v1} and ϑ_{w1} , a unique set of ϑ_{v2} , ϑ_{v3} , ϑ_{w2} and ϑ_{w3} can be calculated based on Eq. (26). Lastly, upon inserting the first two of Eqs. (29) into the first of Eqs. (25), we obtain

$$[\mathbf{v}_1]_B = [-\sin(\vartheta_{v1} + \psi), \cos(\vartheta_{v1} + \psi), 0]^T \quad (46)$$

from which a unique ψ can be determined for each value of ϑ_{v1} . To this end, $\{\mathbf{w}_i\}_1^3$, as displayed in Eq. (25), are obtained, thereby resolving the FK problem. It is worth mentioning that the characteristic equations in $\tan(\vartheta_{s2}/2)$ or $\tan(\vartheta_{s3}/2)$ can also be obtained simply upon exchanging subscripts 1, 2, 3 with 2, 3, 1 or 3, 1, 2 in the aforementioned formulations.

Based on the previous derivations, up to four distinct values of T can be obtained, seemingly resulting in up to $4 \times 2 = 8$ distinct FK solutions. However, numerical studies suggest that at most two of the solutions for T are valid, while the remaining ones either involve complex numbers or lead to $\sin \phi > 1$ or $\sin \phi < -1$. After discarding the invalid solutions, up to $2 \times 2 = 4$ distinct sets of valid FK solutions can be obtained in the numerical studies. This result aligns with the conclusion we obtained in Appendix A, namely, the robot class under study admits at most four distinct assembly modes.

It is worth noting that slight modifications need to be incorporated into the aforementioned procedure in the special case where $\beta_1 = \beta_2 = \beta_3 = \pi/2$. This particular scenario will be discussed in Section 3.5.

3.4. The fourth formulation for general 3-RRR SPMs of the robot class under study

Next, the fourth formulation is presented, which utilizes the same set of parameters and constraint equations as the third formulation. However, a different procedure for the elimination of variables is proposed, ultimately resulting in a quartic univariate characteristic equation in $t \equiv 1 - \sin \phi$.

Firstly, Eqs. (32) is rearranged into the following form:

$$(1 + \sin \phi) \cos \vartheta_{mi} + (1 - \sin \phi) \cos \vartheta_{si} = 2 \cos \beta_i, \quad \text{for } i = 1, 2, 3 \quad (47)$$

Moreover, upon resorting to the last two of Eqs. (35), ϑ_{s2} , ϑ_{s3} , ϑ_{m2} and ϑ_{m3} can be expressed in terms of ϑ_{s1} and ϑ_{m1} , leading to

$$\begin{aligned} (1 + \sin \phi) \cos \vartheta_{m1} &= 2 \cos \beta_1 - (1 - \sin \phi) \cos \vartheta_{s1} \\ (1 + \sin \phi) \cos \mu_{12} \cos \vartheta_{m1} - (1 + \sin \phi) \sin \mu_{12} \sin \vartheta_{m1} &= 2 \cos \beta_2 - (1 - \sin \phi) \cos(\vartheta_{s1} + \sigma_{12}) \\ (1 + \sin \phi) \cos \mu_{13} \cos \vartheta_{m1} - (1 + \sin \phi) \sin \mu_{13} \sin \vartheta_{m1} &= 2 \cos \beta_3 - (1 - \sin \phi) \cos(\vartheta_{s1} + \sigma_{13}) \end{aligned} \quad (48)$$

Then, $\cos \vartheta_{m1}$ can be calculated from the first of Eq. (48), namely,

$$\cos \vartheta_{m1} = \frac{2 \cos \beta_1 - (1 - \sin \phi) \cos \vartheta_{s1}}{1 + \sin \phi} \quad (49)$$

which, after plugging into the last two of Eqs. (48), yields

$$\begin{aligned} (1 + \sin \phi) \sin \mu_{12} \sin \vartheta_{m1} &= 2(\cos \beta_1 \cos \mu_{12} - \cos \beta_2) + (1 - \sin \phi)(\cos(\vartheta_{s1} + \sigma_{12}) - \cos \vartheta_{s1} \cos \mu_{12}) \\ (1 + \sin \phi) \sin \mu_{13} \sin \vartheta_{m1} &= 2(\cos \beta_1 \cos \mu_{13} - \cos \beta_3) + (1 - \sin \phi)(\cos(\vartheta_{s1} + \sigma_{13}) - \cos \vartheta_{s1} \cos \mu_{13}) \end{aligned} \quad (50)$$

Next, a constraint equation only involving ϑ_{s1} and $t \equiv 1 - \sin \phi$ can be obtained upon eliminating $\sin \vartheta_{m1}$ from Eqs. (50), namely,

$$k_1 t \cos \vartheta_{s1} + k_2 t \sin \vartheta_{s1} + k_3 = 0 \quad (51)$$

with

$$\begin{aligned} k_1 &\equiv \sin \mu_{13} \cos \sigma_{12} - \sin \mu_{12} \cos \sigma_{13} - \sin \mu_{23}, \quad k_2 \equiv \sin \mu_{12} \sin \sigma_{13} - \sin \mu_{13} \sin \sigma_{12} \\ k_3 &\equiv 2 \cos \beta_3 \sin \mu_{12} + 2 \cos \beta_1 \sin \mu_{23} + 2 \cos \beta_2 \sin \mu_{13} \end{aligned} \quad (52)$$

Moreover, upon solving for $\sin \vartheta_{m1}$ from the first of Eqs. (50), and plugging its expression, together with Eq. (49), into the relation $\cos^2 \vartheta_{m1} + \sin^2 \vartheta_{m1} = 1$, a second constraint equation only involving ϑ_{s1} and t is obtained, namely,

$$k_4 t^2 \cos^2 \vartheta_{s1} + k_5 t^2 \cos \vartheta_{s1} \sin \vartheta_{s1} + k_6 t \cos \vartheta_{s1} + k_7 t \sin \vartheta_{s1} + (k_8 t^2 + k_9 t + k_{10}) = 0 \quad (53)$$

with

$$\begin{aligned} k_4 &\equiv 2 \cos \sigma_{12} (\cos \sigma_{12} - \cos \mu_{12}), \quad k_5 \equiv 2 \sin \sigma_{12} (\cos \mu_{12} - \cos \sigma_{12}) \\ k_6 &\equiv 4 \cos \beta_2 (\cos \mu_{12} - \cos \sigma_{12}) + 4 \cos \beta_1 (\cos \mu_{12} \cos \sigma_{12} - 1) \\ k_7 &\equiv 4 \sin \sigma_{12} (\cos \beta_2 - \cos \beta_1 \cos \mu_{12}), \quad k_8 \equiv (\cos 2\mu_{12} - \cos 2\sigma_{12})/2 \\ k_9 &\equiv 4 \sin^2 \mu_{12}, \quad k_{10} \equiv 2(2 \cos^2 \beta_2 + \cos 2\beta_1 + \cos 2\mu_{12} - 4 \cos \beta_1 \cos \beta_2 \cos \mu_{12}) \end{aligned} \quad (54)$$

To this end, a univariate equation in t can be obtained once ϑ_{s1} is eliminated from Eqs. (51) and (53). To achieve this, we first convert Eq. (51) into the following form:

$$\sin \vartheta_{s1} = -(k_1 t \cos \vartheta_{s1} + k_3)/(k_2 t) \quad (55)$$

Then, upon substituting Eq. (55) into Eq. (53), a quadratic equation in $\cos \vartheta_{s1}$ can be obtained, with t being the only unknown involved in its three coefficients, namely,

$$(k_2 k_4 - k_1 k_5) t^2 \cos^2 \vartheta_{s1} + (k_2 k_6 - k_3 k_5 - k_1 k_7) t \cos \vartheta_{s1} + k_2 k_8 t^2 + k_2 k_9 t - k_3 k_7 + k_2 k_{10} = 0 \quad (56)$$

Moreover, upon inserting Eq. (55) into the relation $\cos^2 \vartheta_{s1} + \sin^2 \vartheta_{s1} = 1$, a second quadratic equation in $\cos \vartheta_{s1}$ can be obtained, with t being the only unknown involved in its three coefficients, namely,

$$(k_1^2 + k_2^2) t^2 \cos^2 \vartheta_{s1} + 2k_1 k_3 t \cos \vartheta_{s1} - (k_2 t - k_3)(k_2 t + k_3) = 0 \quad (57)$$

Finally, $\cos \vartheta_{s1}$ can be eliminated based on the following procedure, thereby resulting in a univariate polynomial in t : Let us regard Eqs. (56) and (57) as two linear equations in $\cos^2 \vartheta_{s1}$ and $\cos \vartheta_{s1}$, from which $\cos^2 \vartheta_{s1}$ and $\cos \vartheta_{s1}$ can be computed as

$$\cos \vartheta_{s1} = \frac{\lambda_1 t^2 + \lambda_2 t + \lambda_3}{\lambda_7 t}, \quad \cos^2 \vartheta_{s1} = \frac{\lambda_4 t^2 + \lambda_5 t + \lambda_6}{\lambda_7 t^2} \quad (58)$$

with

$$\begin{aligned} \lambda_1 &\equiv k_2^3 k_4 + k_2^3 k_8 - k_1 k_2^2 k_5 + k_1^2 k_2 k_8, & \lambda_2 &\equiv k_2^3 k_9 + k_1^2 k_2 k_9 \\ \lambda_3 &\equiv k_2^3 k_{10} - k_3 k_2^2 k_7 + k_1^2 k_2 k_{10} - k_2 k_3^2 k_4 + k_1 k_3^2 k_5 - k_1^2 k_3 k_7, & \lambda_4 &\equiv -k_2^3 k_6 + k_3 k_2^2 k_5 + k_1 k_2^2 k_7 - 2k_1 k_2 k_3 k_8 \\ \lambda_5 &\equiv -2k_1 k_2 k_3 k_9, & \lambda_6 &\equiv k_2 k_3^2 k_6 + k_1 k_3^2 k_7 - 2k_1 k_2 k_3 k_{10} - k_3^3 k_5 \\ \lambda_7 &\equiv k_2^2 (k_3 k_5 - k_2 k_6) - k_1^2 (k_3 k_5 + k_2 k_6) + k_1 k_2 (2k_3 k_4 + k_2 k_7) + k_1^3 k_7 \end{aligned} \quad (59)$$

Then, based on the equality between the second of Eqs. (58) and the square of the first of Eqs. (58), a quartic univariate characteristic equation in t can be established, namely,

$$\lambda_2^2 t^4 + 2\lambda_1 \lambda_2 t^3 + (\lambda_2^2 + 2\lambda_1 \lambda_3 - \lambda_4 \lambda_7) t^2 + (2\lambda_2 \lambda_3 - \lambda_5 \lambda_7) t + \lambda_3^2 - \lambda_6 \lambda_7 = 0 \quad (60)$$

from which up to four solutions for t can be obtained. Moreover, for each t thus obtained, a unique set of $\sin \phi$ and ϕ can be determined based on the relation $t = 1 - \sin \phi$ and the condition $\phi \in [-\pi/2, \pi/2]$, after which a unique set of $\cos \vartheta_{s1}$, $\sin \vartheta_{s1}$, $\cos \vartheta_{m1}$ and $\sin \vartheta_{m1}$ can be determined based on Eqs. (58), (55), (49) and (50). After that, two sets of ϑ_{v1} , ϑ_{w1} and ψ can be determined for each of t thus obtained, using exactly the same procedure that we utilized in the third formulation.

Similar to the case for the third formulation, the previous derivations appear to yield up to 4×2 distinct sets of FK solutions. However, numerical studies once again suggest that at most two of the solutions for t are valid, while the remaining ones either involve complex numbers or lead to $\sin \phi > 1$ or $\sin \phi < -1$. After discarding the invalid solutions, up to $2 \times 2 = 4$ distinct sets of valid FK solutions can be obtained.

3.4.1. Remarks on the fourth formulation

Firstly, it is worth mentioning that, although the aforementioned formulation is derived with $\{\vartheta_{m1}, \vartheta_{s1}\}$ as intermediate variables, similar formulations can be derived with the aid of $\{\vartheta_{m2}, \vartheta_{s2}\}$ or $\{\vartheta_{m3}, \vartheta_{s3}\}$; this can be achieved simply upon exchanging subscripts 1, 2, 3 with 2, 3, 1 or 3, 1, 2 in the aforementioned formulations. Next, we discuss two special cases under which certain modifications need to be incorporated into the aforementioned formulation.

The first case arises when all the coefficients in the characteristic equation (Eq. (60)) vanish. Fortunately, in most cases, this issue can be resolved by employing those alternative formulations derived with the aid of $\{\vartheta_{m2}, \vartheta_{s2}\}$ or $\{\vartheta_{m3}, \vartheta_{s3}\}$. However, this strategy fails in a special sub-case where $\theta_{12} = \pm \gamma_3 + k_1 \pi$ and $\theta_{13} = \mp \gamma_2 + k_2 \pi$, with $k_1, k_2 \in \mathbb{Z}$. More specifically,

- When $\theta_{12} = \gamma_3 + k_1 \pi$ and $\theta_{13} = -\gamma_2 + k_2 \pi$, it can be readily verified that $\sin \sigma_{12} = \sin \sigma_{13} = \sin \sigma_{23} = 0$. In this case, θ_{s1} can be directly eliminated from Eqs. (48), yielding a system of two constraint equations in ϑ_{m1} and t . After that, a quadratic equation in t can be readily obtained upon eliminating ϑ_{m1} .
- When $\theta_{12} = -\gamma_3 + k_1 \pi$ and $\theta_{13} = \gamma_2 + k_2 \pi$, it can be readily verified that $\sin \mu_{12} = \sin \mu_{13} = \sin \mu_{23} = 0$. In this case, Eqs. (50) simplifies to a system of two constraint equations in ϑ_{s1} and t , which can be converted to a quadratic equation in t upon eliminating ϑ_{s1} . After that, the solutions for t can be readily obtained.

Once t is available, $\{\vartheta_{s1}, \vartheta_{m1}\}$ can be determined from Eqs. (48). After that, the remaining unknowns can be calculated using the procedure introduced in Section 3.3.

The second case arises when $\lambda_7 = 0$. Although t can still be calculated as usual, determining $\cos \vartheta_{s1}$ from Eq. (58) becomes impossible as all the denominators therein vanish in this scenario. Fortunately, $\cos \vartheta_{s1}$ can be obtained by resorting to Eq. (56) or Eq. (57). After that, the remaining unknowns can be calculated using the procedure introduced in Section 3.3.

3.5. A subclass admitting quadratic characteristic polynomials

Next, we focus on a special subclass of the robot class under study, featuring three 90° distal links. As will be shown shortly, the FK of this subclass, when using the four aforementioned formulations, takes on a simpler form, all yielding quadratic characteristic equations with simpler coefficient expressions.

Firstly, the characteristic equation in the first formulation, as displayed in Eq. (11), can be further simplified for this subclass. Indeed, η_2, η_4 and η_6 all vanish in this case, while $\eta_1, \eta_3, \eta_5, \eta_7$ and η_8 simplify to

$$\begin{aligned}\eta_1 &\equiv -m_1 \sin \theta_{13} / (m_2 \sin \theta_{23}), & \eta_3 &\equiv -m_1 \sin \theta_{12} / (m_3 \sin \theta_{32}), & \eta_5 &\equiv m_1^2 (\sin^2 \theta_{12} + \sin^2 \theta_{13} - \sin^2 \theta_{23}) \\ \eta_7 &\equiv (m_1^2 - m_2^2 - m_3^2) \sin^2 \theta_{23}, & \eta_8 &\equiv 2m_2 m_3 \sin^2 \theta_{23}\end{aligned}\quad (61)$$

Accordingly, the characteristic equation shown in Eq. (11) simplifies to

$$\zeta_4 \sin^4 \epsilon_1 + \zeta_2 \sin^2 \epsilon_1 + \zeta_0 = 0 \quad (62)$$

with

$$\begin{aligned}\zeta_4 &= -4m_1^4 \sin^2 \theta_{12} \sin^2 \theta_{13} \\ \zeta_2 &= m_1^2 (m_1^2 + m_2^2 + m_3^2 - (m_1^2 + m_2^2 - m_3^2) \cos(2\theta_{12}) - (m_1^2 - m_2^2 + m_3^2) \cos(2\theta_{13}) + (m_1^2 - m_2^2 - m_3^2) \cos(2\theta_{23})) \\ \zeta_0 &= (m_1 - m_2 - m_3)(m_1 + m_2 - m_3)(m_1 - m_2 + m_3)(m_1 + m_2 + m_3) \sin^2 \theta_{23}\end{aligned}\quad (63)$$

Notice that a common factor $\sin^2 \theta_{23}$ has been eliminated in the above expressions of ζ_0, ζ_2 and ζ_4 for the sake of simplicity. Interestingly, the characteristic equation, as per Eq. (62), becomes a quadratic equation in $\sin^2 \epsilon_1$, resulting in up to two distinct solutions for $\sin^2 \epsilon_1$ or, equivalently, four distinct solutions for $\sin \epsilon_1$. Afterwards, the identification of the valid solutions for $\sin \epsilon_1$, along with the determination of the remaining unknowns, can be carried out using the procedures introduced in Sections 3.1 or 3.2. Interestingly, this special subclass also admits up to four distinct FK solutions, which is consistent with the general case.

Secondly, the characteristic equation in the second formulation, as displayed in Eq. (21), can also be further simplified; indeed, δ_2, δ_5 and δ_7 all vanish in this case, while $\delta_1, \delta_3, \delta_4, \delta_6$ and δ_8 reduce to

$$\delta_1 \equiv 1, \quad \delta_3 \equiv \cos \theta_{12}, \quad \delta_4 \equiv \cos \gamma_3, \quad \delta_6 \equiv -m_2 \sin \theta_{23}, \quad \delta_8 \equiv m_1 \sin \theta_{13} \quad (64)$$

Accordingly, the characteristic equation shown in Eq. (21) simplifies to

$$\zeta'_4 \sin^4 \epsilon_1 + \zeta'_2 \sin^2 \epsilon_1 + \zeta'_0 = 0 \quad (65)$$

with

$$\zeta'_4 = -m_1^2 \sin^2 \theta_{12} \sin^2 \theta_{13}, \quad \zeta'_2 = m_1^2 \sin^2 \theta_{13} + m_2^2 \sin^2 \theta_{23} + 2m_1 m_2 m_4 \cos \theta_{12} \sin \theta_{13} \sin \theta_{23}, \quad \zeta'_0 = -m_2^2 m_3^2 \sin^2 \theta_{23} \quad (66)$$

which also becomes a quadratic characteristic equation in $\sin^2 \epsilon_1$, from which up to four distinct solutions for $\sin \epsilon_1$ can be obtained. After that, the identification of the valid solutions for $\sin \epsilon_1$, along with the determination of the remaining unknowns, can also be carried out using the procedures introduced in Sections 3.1 or 3.2.

Thirdly, the characteristic equation in the third formulation, as displayed in Eq. (42), can be simplified to the following form:

$$(\lambda_5 T^2 - 2\lambda_6 T - \lambda_5)^2 = 0 \quad (67)$$

with

$$\lambda_5 \equiv -\sin \mu_{12} \cos \mu_{13} + \sin \mu_{12} \cos \sigma_{13} + \sin \mu_{13} (\cos \mu_{12} - \cos \sigma_{12}), \quad \lambda_6 \equiv \sin \mu_{13} \sin \sigma_{12} - \sin \mu_{12} \sin \sigma_{13} \quad (68)$$

or, equivalently,

$$\begin{bmatrix} \lambda_5 \\ \lambda_6 \end{bmatrix} = \begin{bmatrix} -2 \sin \gamma_1 & -2 \sin \gamma_2 \sin \gamma_3 & 2 \sin \gamma_2 \sin \gamma_3 \\ 0 & 4 \sin \gamma_2 \cos \gamma_3 & 4 \cos \gamma_2 \sin \gamma_3 \end{bmatrix} \begin{bmatrix} \sin \theta_{12} \sin \theta_{13} \\ \sin \theta_{12} \cos \theta_{13} \\ \cos \theta_{12} \sin \theta_{13} \end{bmatrix} \quad (69)$$

Apparently, Eq. (67) simplifies to a quadratic characteristic equation in T , namely,

$$\lambda_5 T^2 - 2\lambda_6 T - \lambda_5 = 0 \quad (70)$$

which admits up to two distinct solutions for T . Moreover, similar to the general case, a unique ϑ_{s1} can be determined via Eq. (41) for each T thus obtained. However, ϑ_{m1} can no longer be determined by Eq. (38) in this case, as the two equations therein become linearly dependent. As a result, one needs to resort to the first of Eqs. (36) and the identity $\cos^2 \vartheta_{m1} + \sin^2 \vartheta_{m1} = 1$, from which two possible solutions for ϑ_{m1} can be obtained for each ϑ_{s1} thus obtained. This leads to up to four distinct sets of solutions for $\{T, \vartheta_{s1}, \vartheta_{m1}\}$, after which a unique $\sin \phi$ can be determined by Eq. (33) for each set. Finally, the identification of the valid solutions for $T, \vartheta_{s1}, \vartheta_{m1}$ and the determination of the remaining unknowns can be carried out using the procedures introduced in Section 3.3.

Lastly, the characteristic equation in the fourth formulation, as displayed in Eq. (60), can also be further simplified. Indeed, $k_3 = k_6 = k_7 = 0$ in this case and k_{10} simplifies to $-k_9$; accordingly, Eq. (60) simplifies to

$$(\lambda_1 t^2 + \lambda_2 t - \lambda_2)^2 = 0 \quad (71)$$

with

$$\begin{aligned}\lambda_1 &= (\cos \mu_{12} - \cos \sigma_{12})(\cos \sigma_{13} - \cos \mu_{13})(\cos \sigma_{23} - \cos \mu_{23}) \\ \lambda_2 &= 2(\sin^2 \mu_{12} + \sin^2 \mu_{13} + \sin^2 \mu_{23}) - 4 \sin \mu_{13} \sin \mu_{12} \cos \sigma_{23} + 4 \sin \mu_{23} \sin \mu_{12} \cos \sigma_{13} \\ &\quad - 4 \sin \mu_{13} \sin \mu_{23} \cos \sigma_{12}\end{aligned}\quad (72)$$

or, equivalently,

$$\begin{aligned}\lambda_1 &= 8 \sin \gamma_1 \sin \gamma_2 \sin \gamma_3 \sin \theta_{12} \sin \theta_{13} \sin \theta_{23} \\ \lambda_2 &= 4 \sin \gamma_2 \sin \gamma_3 \cos \gamma_1 \cos(2\theta_{23}) + 4 \sin \gamma_1 \sin \gamma_2 \cos \gamma_3 \cos(2\theta_{12}) + 4 \sin \gamma_1 \sin \gamma_3 \cos \gamma_2 \cos(2\theta_{13}) \\ &\quad + 2 \sin^2 \gamma_1 + 2 \sin^2 \gamma_2 + 2 \sin^2 \gamma_3 - 2\lambda_1\end{aligned}\quad (73)$$

Notice that a common factor $2 \sin^2 \mu_{12} (\sin \mu_{12} \sin \sigma_{13} - \sin \mu_{13} \sin \sigma_{12})$ has been eliminated in the expressions of $\{\lambda_i\}_1^2$ in Eqs. (72) and (73). To this end, a quadratic characteristic equation in t is obtained, namely,

$$\lambda_1 t^2 + \lambda_2 t - \lambda_2 = 0 \quad (74)$$

which admits up to two distinct solutions for t ; after that, a unique $\sin \phi$ can be obtained for each t thus obtained. However, numerical studies suggest that at most one of these solutions is valid, while the other one needs to be discarded as it either involves complex numbers or yields $\sin \phi > 1$ or $\sin \phi < -1$. Following this, two possible solutions for $\cos \vartheta_{s1}$ can be obtained by resorting to Eqs. (56) or (57). Finally, the remaining unknowns can be determined using the procedures introduced in Section 3.4.

3.6. Potential applications of the proposed formulations to certain PMs with alternative topologies

It is worth mentioning that the four formulations proposed in this paper are not only applicable to the robot class under study, but also to a variety of PMs with alternative topologies. For instance, various SPMs have been proposed, featuring topologies such as 3-RRRR, 3-RRRRR, 3-PRRR, 3-RPRR, 3-RRPR, and 3-RRRP [45–47], as well as others like 3-RRS [36], 3-RSR [48], 3-RCC [49], 3-URC [50], 3-RUU [47], and 3-CPU [51]. It can be readily seen from these studies that these PMs are kinematically equivalent [36,48,49,52] to 3-RRR SPMs, i.e., their MPs will attain the same pose when functioning in the same assembly mode with an identical set of input joint variables.² These alternative topologies have been proposed for a variety of purposes, including minimizing internal forces, reducing link interference, expanding workspace volume, enabling reconfigurability, etc. [35,53,54]. Apparently, the proposed formulations are also applicable to these SPMs, provided that their relevant joint axes are arranged such that their 3-RRR kinematic equivalents meet the conditions introduced in Section 2. Fig. 5(a)–(c) present three such examples, all kinematically equivalent to the 3-RRR SPM shown in Fig. 2(b), with topologies of 3-RCC, 3-RSR, and 3-URC, respectively. To better understand this equivalence, let us consider the 3-RCC SPM shown in Fig. 5(a), which can be regarded as a variation of the 3-RRR SPM shown in Fig. 2(b) by replacing all six passive R joints with passive C joints. It can be readily verified that the number of degrees of freedom of this mechanism remains three [55], and the C joints will not experience any sliding motion when all nine joint axes intersect perfectly at a common point (as is the current case), thereby justifying the aforementioned kinematic equivalence. It is noteworthy, however, that the introduction of sliding motion capabilities into the limbs is not without benefit. In fact, the additional sliding DOFs help reduce the number of overconstraints [55], allowing the mechanism to remain movable even when the nine axes do not intersect perfectly at a common point. Such a technique for alleviating overconstraints has been successfully applied in real-world applications—see, for example, the well-known Agile Wrist with a 3-RCC topology [55], which reduces machining and assembly difficulties. Similarly, it can be easily verified that the other two examples shown in Figs. 5(b) and 5(c) are also kinematically equivalent to the 3-RRR SPM shown in Fig. 2(b). For further examples and a more detailed explanation of kinematic equivalence, please refer to the supplementary material.

Additionally, it is worth noting that there are certain other types of PMs whose orientation problems are kinematically equivalent to those of 3-RRR SPMs—see, for example, the orientation problems of the 3-CCC [56], 3-CR(Pa)(Pa)R [32], 3-CPPRR [57], 3-P(Pa)(Pa)RRR [58] and 3-RRRRRR + PPPS [59] parallel mechanisms. As a result, the proposed formulations can also be applied to these PMs, provided that their geometry is specified to ensure that their corresponding 3-RRR kinematic equivalents belong to the robot class under study.

4. Case study

In order to validate the different formulations introduced in the previous section, a large number of case studies have been conducted for the four design examples depicted in Fig. 2, with their architectural parameters listed in Table 1. It is noteworthy that these four design examples have been carefully selected to represent cases ranging from the most general to the most symmetric geometry within the robot class. Moreover, these examples also include instances from the special subclass discussed in Section 3.5, ensuring they are sufficiently representative to demonstrate the effectiveness of the proposed formulations across different geometries. For clarity, we will present only one set of these case studies below, using a common set of input joint variables,

² Note that this holds under the condition that their robot configurations lie within their respective feasible workspaces and are not subject to singularities.

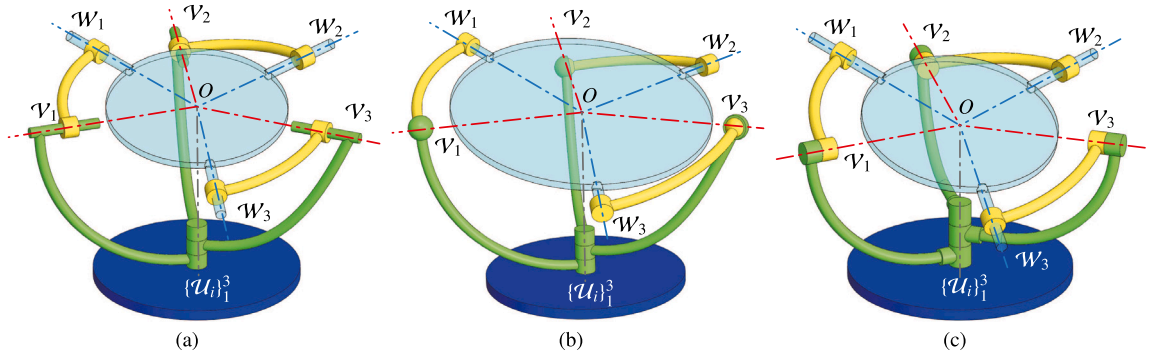


Fig. 5. Examples of three SPMs that are kinematically equivalent to the 3-RRR SPM shown in Fig. 2(b), each with a topology of (a) 3-RCC, (b) 3-RSR, and (c) 3-URC.

arbitrarily chosen as $\theta = [0.9, 3.2, 5.2]^T$. Additionally, the formulation proposed by Gosselin et al. [60] has also been evaluated for cross-validation of the results obtained from the four proposed formulations.

For the first design example depicted in Fig. 2(a), the characteristic equations for the four proposed formulations, as given by Eqs. (11), (21), (42) and (60), as well as that for the formulation used for cross-validation [60], can be evaluated as follows:

$$\begin{aligned}
 &-0.816738 \sin^4 \epsilon_1 - 0.0973565 \sin^3 \epsilon_1 + 1.60278 \sin^2 \epsilon_1 + 0.120669 \sin \epsilon_1 - 0.603609 = 0 \\
 &-0.224843 \sin^4 \epsilon_1 - 0.0268017 \sin^3 \epsilon_1 + 0.441235 \sin^2 \epsilon_1 + 0.0332194 \sin \epsilon_1 - 0.16617 = 0 \\
 &-9.3494T^4 - 0.0759335T^3 + 9.68163T^2 + 0.0513714T - 2.43755 = 0 \\
 &0.000520185t^4 - 0.00424632t^3 + 0.0127647t^2 - 0.0167304t + 0.00807507 = 0 \\
 &-0.402177r^8 + 0.1608r^7 + 2.66293r^6 - 0.0365391r^5 - 2.5767r^4 - 0.0365391r^3 \\
 &\quad + 2.66293r^2 + 0.1608r - 0.402177 = 0
 \end{aligned} \tag{75}$$

whose solutions are found to be³

$$\begin{aligned}
 \sin \epsilon_1 &= -1.2213, -0.75933, 0.66753, 1.1939 \\
 T &= -0.77509, -0.66312, 0.65153, 0.77856 \\
 t &= 2.4752, 1.59052, 1.54471, 2.5527 \\
 r &= -2.1739, -0.46000, 0.38263, 2.6135, -0.81882 - 0.57405i, -0.81882 + 0.57405i, \\
 &\quad 0.83761 - 0.54628i, 0.83761 + 0.54628i
 \end{aligned} \tag{76}$$

with $r \equiv \tan(\epsilon_1/2)$. Similarly, for the second design example depicted in Fig. 2(b), the characteristic equations for all five formulations can be evaluated as follows:

$$\begin{aligned}
 &-0.816738 \sin^4 \epsilon_1 - 0.0102109 \sin^3 \epsilon_1 + 1.29067 \sin^2 \epsilon_1 + 0.0100889 \sin \epsilon_1 - 0.458454 = 0 \\
 &-0.190972 \sin^4 \epsilon_1 - 0.00238753 \sin^3 \epsilon_1 + 0.301788 \sin^2 \epsilon_1 + 0.00235901 \sin \epsilon_1 - 0.107197 = 0 \\
 &-12.5612T^4 - 0.0629368T^3 + 8.41814T^2 + 0.0231779T - 1.37296 = 0 \\
 &0.000520185t^4 - 0.00424632t^3 + 0.0129335t^2 - 0.0174189t + 0.00875337 = 0 \\
 &-0.259446r^8 + 0.0114189r^7 + 1.88385r^6 - 0.0119711r^5 - 3.10867r^4 - 0.0119711r^3 \\
 &\quad + 1.88385r^2 + 0.0114189r - 0.259446 = 0
 \end{aligned} \tag{77}$$

whose solutions are found to be

$$\begin{aligned}
 \sin \epsilon_1 &= -1.0210, -0.73987, 0.72869, 1.0197 \\
 T &= -0.62474, -0.53162, 0.52760, 0.62376 \\
 t &= 2.2860, 1.79417, 1.78695, 2.2960 \\
 r &= -2.2609, -0.44231, 0.43250, 2.3122, -0.97940 - 0.20191i, -0.97940 + 0.20191i, \\
 &\quad 0.98067 - 0.19565i, 0.98067 + 0.19565i
 \end{aligned} \tag{78}$$

³ In the following case studies, the solutions for $\sin \epsilon_1$, as obtained via the second formulation, are not displayed since they are exactly the same as those obtained from the first formulation.

Table 2

FK Solutions for the four design examples shown in Fig. 2.

sol	Design Example I				Design Example II				Design Example III				Design Example IV			
	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
ϵ_1	-0.86229	-2.2793	0.73088	2.4107	-0.83288	-2.3087	0.81641	2.3252	-1.5243	-1.6173	0.92142	2.2202	-1.2381	-1.9035	1.2381	1.9035
ϵ_2	-2.5955	-0.54611	1.7304	1.4112	-2.2700	-0.87160	1.9805	1.1611	-2.3237	-0.81786	1.2152	1.9264	-1.2607	-1.8809	1.2607	1.8809
ϵ_3	-1.3470	-1.7946	2.4963	0.64534	-1.7351	-1.4065	2.1090	1.0326	-0.91273	-2.2289	2.6138	0.52777	-2.2547	-0.88689	2.2547	0.88689
θ_{E1}	-0.39809	2.7435	0.27776	3.4194	0.30524	3.4468	-0.32045	2.8211	-1.0079	2.1337	0.56730	3.7089	-2.0085	1.1331	1.1331	4.2747
θ_{E1}	-0.77299	2.3686	0.87714	4.0187	-1.2825	1.8591	1.2914	4.4330	-0.055944	3.0856	0.65619	3.7978	0.54319	3.6848	0.54319	3.6848
ψ	-2.0727	1.0689	-2.7486	0.39303	-2.7760	0.36556	-2.1503	0.99125	-1.4629	1.6787	-3.0381	0.10350	-0.46234	2.6793	2.6793	-0.46234
ϕ	-0.63170	-0.63170	-0.57604	-0.57604	-0.91763	-0.91763	-0.90585	-0.90585	-0.76722	-0.76722	-0.53877	-0.53877	-0.88673	-0.88673	-0.88673	-0.88673

Moreover, for the third design example depicted in Fig. 2(c), the characteristic equations for all five formulations can be evaluated as follows:

$$\begin{aligned}
& -0.488427 \sin^4 \epsilon_1 - 0.0463714 \sin^3 \epsilon_1 + 1.43554 \sin^2 \epsilon_1 + 0.168029 \sin \epsilon_1 - 0.824489 = 0 \\
& -0.173875 \sin^4 \epsilon_1 - 0.0165078 \sin^3 \epsilon_1 + 0.511037 \sin^2 \epsilon_1 + 0.0598168 \sin \epsilon_1 - 0.29351 = 0 \\
& -11.3119T^4 + 3.46416T^3 + 10.8977T^2 - 1.63414T - 2.67302 = 0 \\
& 2.46978t^4 - 20.0809t^3 + 60.2434t^2 - 79.0007t + 38.2287 = 0 \\
& -0.391347r^8 + 0.159512r^7 + 1.16014r^6 + 0.302452r^5 - 0.606361r^4 + 0.302452r^3 \\
& + 1.16014r^2 + 0.159512r - 0.391347 = 0
\end{aligned} \tag{79}$$

whose solutions are found to be

$$\begin{aligned}
& \sin \epsilon_1 = -1.4038, -0.99892, 0.79646, 1.5114 \\
& T = -0.66611, -0.58849, 0.70152, 0.85931 \\
& t = 2.3147, 1.69414, 1.51308, 2.6087 \\
& r = -1.0476, -0.95455, 0.49633, 2.0148, -0.71233 - 0.70185i, -0.71233 + 0.70185i, \\
& 0.66165 - 0.74981i, 0.66165 + 0.74981i
\end{aligned} \tag{80}$$

Finally, for the fourth design example depicted in Fig. 2(d), which belongs to the special subclass discussed in Section 3.5, the characteristic equations for all four formulations, as given by Eqs. (11), (21), (42) and (60), as well as that for the formulation used for cross-validation [60], can be evaluated as follows:

$$\begin{aligned}
& -0.868315 \sin^4 \epsilon_1 + 2.06709 \sin^2 \epsilon_1 - 1.15363 = 0 \\
& -0.350061 \sin^4 \epsilon_1 + 0.833347 \sin^2 \epsilon_1 - 0.465087 = 0 \\
& 2.54727T^2 - 0.539605T - 2.54727 = 0 \\
& 1.57155t^2 - 6.38886t + 6.38886 = 0 \\
& -0.620116r^8 + 1.96405r^6 - 2.29964r^4 + 1.96405r^2 - 0.620116 = 0
\end{aligned} \tag{81}$$

whose solutions are found to be

$$\begin{aligned}
& \sin \epsilon_1 = -0.94515, -1.2195, 0.94515, 1.2195, \quad T = -0.89968, 1.1115, \quad t = 1.77501, 2.2903 \\
& r = -1.4036, -0.71244, 0.71244, 1.4036, -0.81998 - 0.57239i, -0.81998 + 0.57239i, \\
& 0.81998 - 0.57239i, 0.81998 + 0.57239i
\end{aligned} \tag{82}$$

After discarding the invalid solutions, each of the five formulations yields four sets of valid FK solutions for each design example, which are consistent across all formulations—see Table 2 for the valid solutions and Figs. 6 to 9 for the corresponding configurations.

As stated earlier, we have conducted a large number of case studies for the four design examples shown in Fig. 2, although only one set of these numerical verifications is described in detail for clarity. Among these case studies, three sets are dedicated to the three special cases arising in the formulations derived in Section 3. More specifically, the input joint variables are specified according to the requirements of each of these three special cases, and the relevant results demonstrate that the proposed formulations can effectively handle these cases using the techniques outlined in Section 3. The details of these case studies are omitted from the manuscript due to space limitations, but can be found in the supplementary material.

It is worth mentioning that the FK solutions obtained in all the aforementioned case studies have been verified through inverse kinematics of the corresponding design examples, further confirming the correctness of the proposed formulations. The details of these straightforward verifications are also omitted due to space limitations.

5. Conclusions

In this paper, we investigated a novel class of 3-RRR SPMs with CFFKS, which are characterized by three coplanar intermediate-joint axes and three coplanar distal-joint axes. Notably, five of the architecture parameters can be freely specified for the robot class, indicating that it encompasses a large set of potential designs. The FK of the robot class was formulated using four distinct formulations, all yielding quartic characteristic equations. Moreover, a special subclass of the robot class was investigated, yielding

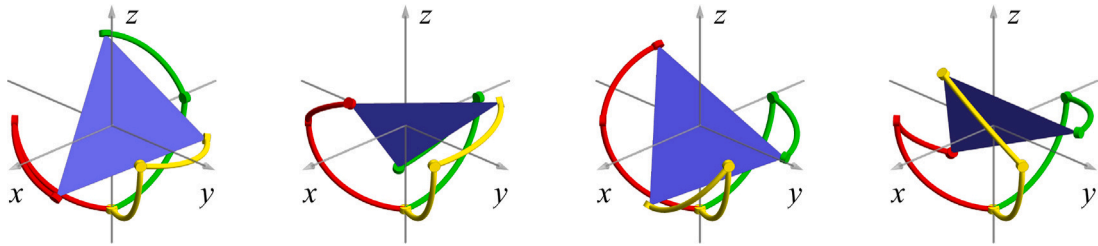


Fig. 6. The configurations corresponding to the FK solutions for Design Example I.

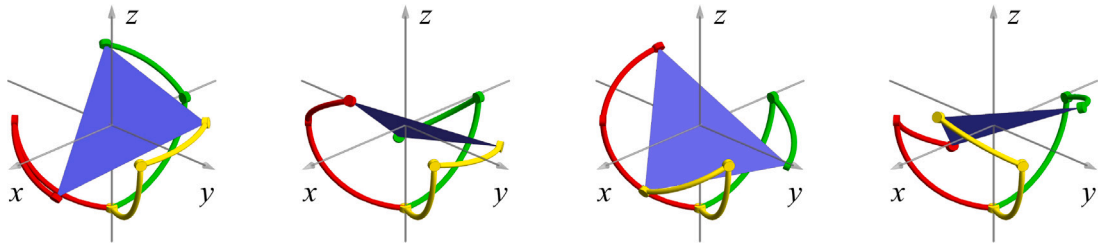


Fig. 7. The configurations corresponding to the FK solutions for Design Example II.

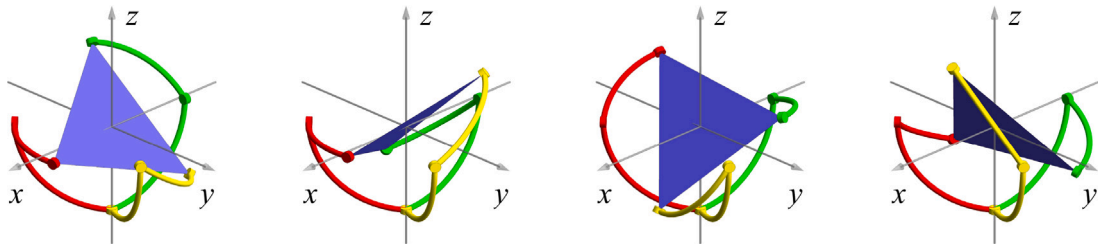


Fig. 8. The configurations corresponding to the FK solutions for Design Example III.

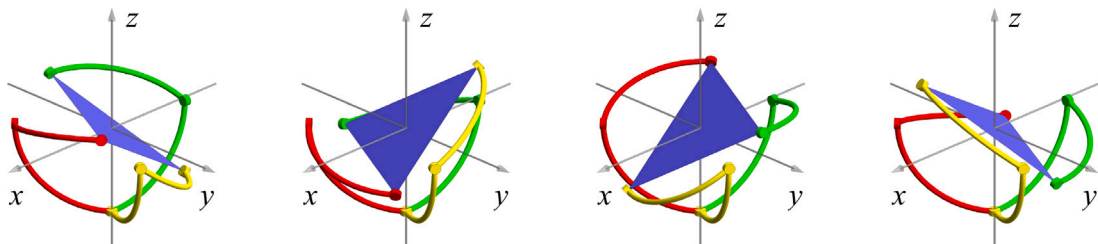


Fig. 9. The configurations corresponding to the FK solutions for Design Example IV.

quadratic characteristic polynomials with simpler coefficient expressions. These formulations not only streamline tasks such as simulation, calibration, and feedback control but also facilitate trajectory planning within the input joint space, including minimal-energy trajectory planning. Next, the maximum number of distinct FK solutions was determined to be four, in contrast to the general case of eight. It is worth mentioning that the robot class under study bears a coaxial layout for the input joint axes, resulting in a number of advantages, such as unlimited rotation capability about certain directions, reduced limb interference, and hence, generally larger workspace volumes. The simple FK relations with CFFKS, together with the aforementioned characteristics, render the robot class potentially attractive in practical applications. Finally, it is worth mentioning that the aforementioned formulations can also find applications in the analysis of certain PMs with alternative topologies.

CRediT authorship contribution statement

Wei Li: Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Shuai Zhang:** Writing – original draft, Methodology, Investigation, Formal analysis,

Conceptualization. **Shenghao Dai**: Validation, Investigation, Formal analysis. **Huaijin Ye**: Visualization, Investigation, Formal analysis. **Weizhong Guo**: Methodology, Investigation, Formal analysis. **Feng Gao**: Writing – review & editing, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Determination of the maximum number of distinct assembly modes

As is well-known, the maximum number of distinct assembly modes for 3-RRR SPMs is eight for the general case, a result that has been demonstrated by various researchers [16–18]. In the subsequent analysis, we determine this number for the robot class under study, which will be shown to be four, in contrast to the general case of eight.

Next, we provide a proof based on the second formulation: As stated in Section 3.2, the quartic characteristic polynomial, as displayed in Eq. (21), admits up to four distinct solutions for $\sin \epsilon_1$; moreover, each valid solution for $\sin \epsilon_1$ corresponds to two distinct values of ϵ_1 , after which a unique combination of ϵ_2 and ϵ_3 can be determined for each ϵ_1 . As a result, the problem at hand becomes proving that at most two distinct valid solutions are present among the aforementioned four solutions for $\sin \epsilon_1$. After that, it follows that the maximum number of distinct valid FK solutions or, equivalently, the maximum number of distinct assembly modes, can be determined to be $2 \times 2 = 4$.

Now, let us introduce a new variable y to represent the characteristic polynomial in the second formulation, i.e., the LHS of the characteristic equation in Eq. (21):

$$y \equiv \zeta'_4 \sin^4 \epsilon_1 + \zeta'_3 \sin^3 \epsilon_1 + \zeta'_2 \sin^2 \epsilon_1 + \zeta'_1 \sin \epsilon_1 + \zeta'_0 \quad (\text{A.1})$$

Moreover, upon plugging Eq. (20) into Eq. (22), the coefficients $\{\zeta'_i\}_1^4$ can be expressed in terms of θ_{12} , θ_{13} , θ_{23} and $\{\beta_i\}_1^3$; specifically, ζ'_4 can be expressed as

$$\zeta'_4 = -m_1^2 \sin^4 \beta_1 \sin^2 \beta_2 \sin^2 \theta_{12} \sin^2 \theta_{13} \quad (\text{A.2})$$

which is always negative unless $\sin \theta_{12}$ or $\sin \theta_{13}$ vanishes.

Now, let us first consider the scenario where $\sin \theta_{12} \neq 0$ and $\sin \theta_{13} \neq 0$. In this case, $\zeta'_4 < 0$, indicating that y approaches negative infinity as “ $\sin \epsilon_1$ approaches positive or negative infinity”.⁴ However, since $\sin \epsilon_1$ must lie within the interval $[-1, 1]$, we can determine the validity of the four possible solutions for $\sin \epsilon_1$ by checking whether they lie within this interval and whether the remaining unknowns (associated with each solution for $\sin \epsilon_1$) have valid values. To do this, let us first look at the number of the solutions for $\sin \epsilon_1$ within the interval $[1, \infty)$. In this vein, we evaluate y and its derivative at $\sin \epsilon_1 = 1$, namely,

$$y_{\sin \epsilon_1=1} = \zeta'_4 + \zeta'_3 + \zeta'_2 + \zeta'_1 + \zeta'_0 = ((\delta_4 + \delta_5)\delta_6 - (\delta_2 + \delta_3)(\delta_7 + \delta_8))^2 \quad (\text{A.3})$$

and

$$\begin{aligned} \dot{y}_{\sin \epsilon_1=1} &= 4\zeta'_4 + 3\zeta'_3 + 2\zeta'_2 + \zeta'_1 \\ &= 2\delta_1^2(\delta_6^2 - (\delta_7 + \delta_8)^2) + 2((\delta_4 + \delta_5)\delta_6 - (\delta_2 + \delta_3)(\delta_7 + \delta_8))(\delta_5\delta_6 - \delta_2\delta_8 - \delta_3\delta_7 - 2\delta_3\delta_8) \end{aligned} \quad (\text{A.4})$$

Apparently, $y_{\sin \epsilon_1=1} \geq 0$; moreover, $\dot{y}_{\sin \epsilon_1=1}$ can be further simplified for the special case that $y_{\sin \epsilon_1=1} = 0$, namely,

$$\dot{y}_{\sin \epsilon_1=1} = 2\delta_1^2(\delta_6^2 - (\delta_7 + \delta_8)^2) = 2\delta_1^2\delta_6^2(1 - \sin^2 \epsilon_2) \quad \text{when } y_{\sin \epsilon_1=1} = 0 \quad (\text{A.5})$$

where the relation shown in the second of Eqs. (18) has been employed to simplify Eq. (A.5). This implies that $\dot{y}_{\sin \epsilon_1=1} \geq 0$ when $y_{\sin \epsilon_1=1} = 0$. However, it is important to note that this deduction relies on the assumption that $\sin^2 \epsilon_2 \leq 1$, which, while true for a valid set of FK solutions, may not hold for an invalid set of FK solutions. Next, we proceed with our discussion by examining the four different cases:

- (a) $y_{\sin \epsilon_1=1} > 0$: Considering that $\zeta'_4 < 0$, it is apparent that there exists at least one root for the quartic characteristic equation within the interval of $(1, \infty)$.
- (b) $y_{\sin \epsilon_1=1} = 0$ and $\dot{y}_{\sin \epsilon_1=1} > 0$: For the same reason as Case (a), there exists at least one root for the quartic characteristic equation within the interval of $(1, \infty)$;
- (c) $y_{\sin \epsilon_1=1} = 0$ and $\dot{y}_{\sin \epsilon_1=1} = 0$: Apparently, the characteristic equation admits a double root at $\sin \epsilon_1 = 1$ in this case.

⁴ It is noteworthy that $\sin \epsilon_1$ is treated as a regular variable in this sentence, whose absolute value can exceed 1.

- (d) $y_{\sin \epsilon_1=1} = 0$ and $\dot{y}_{\sin \epsilon_1=1} < 0$: This seems to lead to a valid solution $\sin \epsilon_1 = 1$; however, as made apparent by Eq. (A.5), this occurs only if the corresponding $\sin^2 \epsilon_2 > 1$, making ϵ_2 a complex number. Thus, the aforementioned solution $\sin \epsilon_1 = 1$ is actually invalid in this case, which must be discarded.

Apparently, all these cases lead to a reduction in the maximum number of valid solutions for $\sin \epsilon_1$ by one.

Next, we examine the number of the solutions within the interval of $(\infty, -1]$. To do this, we evaluate y and its derivative at $\sin \epsilon_1 = -1$, namely,

$$y_{\sin \epsilon_1=-1} = ((\delta_5 - \delta_4)\delta_6 + (\delta_2 - \delta_3)(\delta_7 - \delta_8))^2 \quad (\text{A.6})$$

and

$$\begin{aligned} \dot{y}_{\sin \epsilon_1=-1} &= -4\zeta'_4 + 3\zeta'_3 - 2\zeta'_2 + \zeta'_1 \\ &= -2\delta_1^2(\delta_6^2 - (\delta_7 - \delta_8)^2) - 2((\delta_5 - \delta_4)\delta_6 + (\delta_2 - \delta_3)(\delta_7 - \delta_8))(\delta_5\delta_6 - \delta_2\delta_8 - \delta_3\delta_7 + 2\delta_3\delta_8) \end{aligned} \quad (\text{A.7})$$

Moreover, $\dot{y}_{\sin \epsilon_1=-1}$ can also be further simplified for the special case that $y_{\sin \epsilon_1=-1} = 0$, namely,

$$\dot{y}_{\sin \epsilon_1=-1} = -2\delta_1^2(\delta_6^2 - (\delta_7 - \delta_8)^2) = -2\delta_1^2(\delta_6^2 - \delta_6^2 \sin^2 \epsilon_2) = 2\delta_1^2\delta_6^2(\sin^2 \epsilon_2 - 1) \quad \text{when } y_{\sin \epsilon_1=-1} = 0 \quad (\text{A.8})$$

Now, following a similar reasoning as for the case of the interval $[1, \infty)$, we can draw a similar conclusion for the range $(-\infty, -1]$, which also leads to a reduction in the maximum number of valid solutions for $\sin \epsilon_1$ by one. As a result, the characteristic equation shown in Eq. (21) indeed only admits up to two distinct valid solutions for $\sin \epsilon_1$ when $\sin \theta_{12} \neq 0$ and $\sin \theta_{13} \neq 0$.

Lastly, let us consider the case where $\sin \theta_{12} = 0$ or $\sin \theta_{13} = 0$. It can be readily verified that ζ'_3 also vanishes in this case, making Eq. (21) quadratic. Apparently, at most two distinct solutions for $\sin \epsilon_1$ can be obtained in this scenario, thereby completing the proof.

Appendix B. An alternative routine for the computation of $\{\lambda_i\}_1^7$ in Eqs. (39)

Next, we will streamline the computation of $\{\lambda_i\}_1^7$ by transforming their expressions from Eqs. (39) into a more efficient alternative form. We begin by expressing σ_{12} , σ_{13} , μ_{12} , and μ_{13} , as appearing in Eqs. (39), in terms of θ_{12} , θ_{13} , γ_2 , and γ_3 , with the aid of Eq. (35). Following straightforward manipulations, Eqs. (39) can be reformulated as:

$$\mathbf{A} = \mathbf{K}\boldsymbol{\theta} \quad (\text{B.1})$$

where

$$\begin{aligned} \mathbf{A} &= [\lambda_1 \quad \lambda_2 \quad \lambda_3 \quad \lambda_4 \quad \lambda_5 \quad \lambda_6 \quad \lambda_7]^T \\ \boldsymbol{\theta} &= [\cos \theta_{12} \cos \theta_{13} \quad \cos \theta_{12} \sin \theta_{13} \quad \sin \theta_{12} \cos \theta_{13} \quad \sin \theta_{12} \sin \theta_{13} \quad \cos \theta_{12} \quad \sin \theta_{12} \quad \cos \theta_{13} \quad \sin \theta_{13}]^T \\ \mathbf{K} &= \begin{bmatrix} -k_2 & k_{11} & -k_{11} & k_2 & -k_8 & k_9 & k_6 & -k_7 \\ 0 & 2k_{12} & -2k_{13} & 0 & 0 & 0 & 0 & 0 \\ 0 & -2k_{13} & 2k_{12} & 0 & 0 & -2k_8 & 0 & 2k_6 \\ k_2 & k_{11} & -k_{11} & -k_2 & k_8 & k_9 & -k_6 & -k_7 \\ 0 & -2k_5 & 2k_5 & 2k_1 & 0 & 0 & 0 & 0 \\ 0 & 2k_3 & -2k_4 & 0 & 0 & 0 & 0 & 0 \\ -k_2 & k_{10} & -k_{10} & -k_2 & -k_8 & k_9 & k_6 & -k_7 \end{bmatrix} \end{aligned} \quad (\text{B.2})$$

with

$$\begin{aligned} k_1 &= \sin(\gamma_2 - \gamma_3), \quad k_2 = \cos \beta_1 k_1, \quad k_3 = \cos \gamma_2 \sin \gamma_3, \quad k_4 = \cos \gamma_3 \sin \gamma_2, \quad k_5 = \sin \gamma_2 \sin \gamma_3 \\ k_6 &= \cos \beta_2 \sin \gamma_2, \quad k_7 = \cos \beta_2 \cos \gamma_2, \quad k_8 = \cos \beta_3 \sin \gamma_3, \quad k_9 = \cos \beta_3 \cos \gamma_3, \quad k_{10} = \cos \beta_1 \cos(\gamma_2 - \gamma_3) \\ k_{11} &= \cos \beta_1 \cos(\gamma_2 + \gamma_3), \quad k_{12} = \cos \beta_1 k_3, \quad k_{13} = \cos \beta_1 k_4 \end{aligned} \quad (\text{B.3})$$

Notice that matrix $\mathbf{K}$ is independent of the input joint variables, which thus only needs be calculated once for a given robot, thereby enabling a more efficient routine for the computation of $\{\lambda_i\}_1^7$.

Appendix C. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.mechmachtheory.2025.105930>.

Data availability

No data was used for the research described in the article.

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